

Edge4AV: Is Today's Edge Ready for Tomorrow's Connected and Automated Vehicles?

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Abstract

Connected and automated vehicles (CAVs) need road-user state, infrastructure state, and assistance beyond onboard sensing. However, the presence of vehicle-to-everything (V2X) and fifth-generation (5G) links does not establish that a complete application exchange will finish before its information expires. This readiness question spans two application patterns. Stateless updates can be replaced by newer valid reports, whereas stateful collaborative services need responses that remain correlated to a request or session.

To determine whether today's edge communication is ready for these applications, we design and implement the Intersection Programming Interface (IPI). IPI provides a typed application protocol that combines lightweight local profiles and pre-encoded payloads for selected J2735 message types standardized by SAE International with correlated CAV service operations. We separately evaluate a production Long-Term Evolution cellular V2X (LTE C-V2X) PC5 direct-sidelink path between nearby devices and a private 5G New Radio (NR) Uu vehicle-to-network path through the cellular network on a real autonomous vehicle. The PC5 performance measurements use a custom sequence-matched request/reply probe, whereas the Uu measurements use decoded, accepted IPI acknowledgments. Across both paths, we report all-attempt path-response availability and sender-measured round-trip time (RTT) for application-derived objects under field conditions, shared demand, and service interruption.

The results expose complementary but incomplete operating envelopes. On the measured PC5 path, 2-kibibyte (KiB) reply failure rises from 0.2% to 99.2% as the field path weakens, while all-attempt route path-response availability is 59.1–76.3%. On the measured Uu path, all sender-measured round trips from a request to an accepted acknowledgment (ACK) for a 1- or 2-mebibyte (MiB) object exceed 500 ms. This path response is not a semantically completed service; tighter deadlines or shared demand further contract the usable envelope. Consequently, current direct V2X paths cannot alone support complex CAV applications that need dependable large-object or multi-step exchange. Today's 5G can meet measured necessary communication conditions for some complex CAV exchanges, but usable data size is strictly limited by the application's time budget and operating conditions. Therefore, collaborative CAV applications

need communication methods beyond current V2X and 5G paths used independently and as-is.

1 Introduction

A connected and automated vehicle (CAV) can act only on information that it can observe or receive. Onboard cameras, lidar, and radar provide detailed local observations, but buildings, trucks, vegetation, and other road users can block their view. Moreover, onboard sensors do not reveal a traffic controller's future state, a nearby vehicle's intent, or the result of a service hosted at a nearby network endpoint. Communication extends this observable horizon. Vehicle-to-everything (V2X) links can deliver nearby road-user state and warnings, while cellular links can connect a vehicle to intelligent transportation system (ITS) infrastructure, traffic management, cooperative perception, map services, and remote assistance. Accordingly, standards and research roadmaps position advanced V2X communication as an enabler of cooperative driving, extended sensing, and remote operation [3, 35]. This vision explains why CAVs need communication even as onboard autonomy improves. However, the presence of a radio link does not establish that a complete application exchange will finish before its information expires.

This distinction becomes important because connected-vehicle and CAV applications exchange information in different ways. In this paper, a *stateless update* is an independent report for which a newer valid report can replace an older one. The SAE J2735 message set supports many connected vehicle (CV) and ITS exchanges [37]. For example, a Basic Safety Message (BSM) reports vehicle state, and a Personal Safety Message (PSM) reports vulnerable-road-user state. Map Data (MAP) describes intersection geometry, whereas Signal Phase and Timing (SPaT) reports signal state. These four messages illustrate replaceable state updates. J2735 also defines a Signal Request Message (SRM) and Signal Status Message (SSM), which carry a related signal request and status [20]. By contrast, a *stateful collaborative exchange* belongs to a particular request or session. Perception assistance, planning assistance, map updates, computation offload, and recovery assistance may require a correlated response, a freshness limit, a deadline, progress information, and an explicit failure outcome. Thus, interoperable message content is necessary, but a standalone message does not express the entire lifecycle of a collaborative service.

The two application patterns also place different demands on the network. Periodic signal state is compact and may be common to many receivers. In contrast, intersection guidance combines recurring vehicle-to-infrastructure updates with common signal state and individualized guidance. An emergency maneuver creates an unpredictable bidirectional burst, whereas recovery of a disabled CAV can create a long-lived session. If a help center needs live video, that session becomes uplink-heavy even though its return commands remain small. Sensor-derived objects and map regions add still larger transfers. These flows may coexist, so a deadline-critical request can compete with a sustained stream. This direction imbalance matters because time-division duplex cellular networks allocate different transmission opportunities to the uplink and downlink. Indeed, industry guidance notes that data-producing vertical applications can require uplink capacity and latency that differ from downlink-oriented public-network configurations [34]. Moreover, Third Generation Partnership Project (3GPP) requirements span different payload sizes, rates, latency targets, reliability requirements, and uplink/downlink ratios [3]. A peak data rate or an isolated median delay therefore cannot show whether the network protects the right exchange at the right time.

These differences lead to the central question of this paper: *Can today's deployed V2X and fifth-generation (5G) paths support the protocols, data objects, and time-bounded exchanges that proposed CAV applications actually require?* Here, support means meeting measured necessary communication conditions. It does not mean that a remote service executed correctly or that vehicle safety was established. We answer this question through three checks. First, the interface must preserve message type, source, time, path, session, correlation, expiration, response state, rejection, and termination. Next, the communication path must be evaluated by path-response availability and round-trip time (RTT) across the application's object sizes and operating conditions. Finally, the results must show which application exchanges are practical today and which requirements remain for cellular, radio, edge, CV, and CAV researchers. These checks define readiness at the application boundary. They do not equate readiness with radio-layer peak performance.

Existing technologies provide complementary parts of this interface. Vehicular message standards define interoperable content, while the 3GPP V2X Application Enabler and network-aware middleware provide delivery, resource, or path-selection functions [4, 46]. However, these layers leave application-specific software to connect established ITS messages to the request lifecycle of a time-bounded CAV service. This separation creates an interface gap between message interoperability and complete service semantics.

To close this interface gap, we design the Intersection Programming Interface (IPI). IPI is a typed application protocol

and reference interface for conventional CV messages and CAV service operations. Here, *typed* means that each envelope identifies the kind and structure of its content, and *transport-independent* means that the application contract is not tied to one radio or Internet Protocol transport. IPI provides lightweight typed profiles for BSM, PSM, MAP, SPaT, SRM, and SSM. It also preserves pre-encoded J2735 payloads and adds a cooperative-service message designed for later integration through a regional extension, J2735's mechanism for carrying application-defined content. A session interface supports perception, planning, control, and computation assistance. The common envelope records message, source, intersection, time, and communication-path identifiers. It can also carry session and correlation identifiers. Service messages add a requested time horizon, expiration, payload, confidence, and request, update, completion, or rejection state. Consequently, an ITS client can use familiar signal messages, while a CAV can request a correlated object list or guidance result through the same application interface. IPI is neither a perception algorithm nor a radio technology. Instead, it carries the timestamps, expiration, correlation, and service state needed to evaluate freshness and completion consistently across communication paths.

With that contract, we evaluate two communication paths that are widely used or proposed for vehicular applications. The first is a production, standards-oriented, off-the-shelf Long-Term Evolution cellular V2X (LTE C-V2X) on-board-unit to roadside-unit path. The vendor reports protocol-conformance certification for the two product families. The path uses the PC5 interface, which is the direct sidelink between nearby V2X devices and does not carry the measured packets through a cellular core. The second is an on-site private 5G New Radio (NR) deployment. Its Uu interface, the cellular link between the vehicle gateway and base station, connects the vehicle through provisioned subscriber identity modules, a 5G gateway, the mobile core, and a local edge host. The direct experiment does not measure NR-V2X sidelink. Our cellular scope is an isolated private network with a local application endpoint. Controlled offered load and logical clients stress competing traffic and aggregate demand through one vehicle attachment. Together, the two paths expose path-response behavior for complementary direct and network-assisted communication under controlled field conditions. They do not execute the same application protocol or establish semantic service completion.

The evaluation combines standard vehicular messages with application-derived CAV workloads. These workloads include detector object lists produced by a completed 922-sample V2X-Radar detector run and object sizes derived from public cooperative-driving datasets. We vary payload size, communication direction, transport protocol, field location

and route, obstruction, survey-level 5G radio context, background uplink traffic, and application time budget. We also vary the number of concurrent logical clients and inject receiver or broker restarts. In addition, we test an application-side quality-of-service (QoS) label and check whether captured packets carry a different Internet Protocol traffic marking; they do not. The logical clients share one vehicle host and one 5G attachment. Therefore, that experiment measures aggregate application demand through a shared vehicle path rather than independently scheduled physical vehicles. Most 5G endpoint clocks were not synchronized, so the paper uses sender-measured RTT instead of claiming one-way latency.

The measurements reveal an application-dependent operating envelope. At favorable stationary points, compact direct LTE C-V2X requests receive replies in 99.8–100% of each 1,000-attempt condition. However, across the stationary field-point sequence, the failure rate for a 2-kibibyte (KiB) packet rises from 0.2% to 99.2%. At the next location, all 910 observed attempts fail. Along the driven route, only 59.1–76.3% of attempts receive a reply, although the replies retain low RTT. Thus, measuring delay only for received replies hides route-level unavailability. The 5G path carries larger objects, but object ACK return depends on the application’s time budget. Across the aggregated deadline analysis, 46.6% of 2,000 compact SPaT/state-mirror exchanges miss 100 ms, whereas none miss 500 ms; the state-mirror workload is a request/reply copy of the signal-state payload. Meanwhile, 128–512 KiB exchanges miss 500 ms in 11.64% of 18,000 attempts, and all 14,000 requests carrying 1 or 2 mebibytes (MiB) exceed 500 ms. Controlled offered uplink traffic and service restarts increase tail RTT or remove replies. Location-associated weak-path collections and high logical-client demand are likewise associated with larger tails or lower reply availability on the measured path. Therefore, neither a technology name, a completed transfer, nor a low median is a sufficient readiness test.

This paper makes two contributions:

- **A common protocol for CV, ITS, and CAV applications.** We design and implement IPI as a transport-independent application protocol and reference interface that places selected J2735 profiles and pre-encoded payloads beside a correlated CAV service-operation data model.
- **An application-driven field evaluation.** We characterize the operating envelope of a production LTE C-V2X PC5 path and a private 5G NR Uu path on a real autonomous vehicle. The PC5 product families have vendor-reported conformance certification. We use standard messages and workloads taken from CAV application outputs and datasets.

Together, these contributions determine which application exchanges each path can support within the measured operating conditions. They lead to three insights:

- **Current direct V2X paths cannot, by themselves, support complex CAV applications that require dependable large-object or multi-step service exchange.** Direct V2X remains valuable for compact, fresh local state, whereas larger collaborative objects require a reduced representation or another path.
- **Today’s 5G can support some complex CAV exchanges, but the usable data size is strictly limited by the application’s time budget and operating conditions.** Deadline misses under load, rather than peak downlink rate, set this bound.
- **Collaborative CAV applications need communication methods beyond current V2X and 5G paths used independently and as-is.** Future designs must coordinate application deadlines and service state with network conditions and explicit failure behavior.

The remainder of the paper builds the evidence for these conclusions. We first identify what existing standards, platforms, prototypes, and measurements do and do not establish. We then present IPI and define the application conditions used to judge readiness. Finally, we report the field results, derive the three insights, and translate them into research directions for CV, CAV, ITS, cellular, radio, and edge systems.

2 Related Work

Research on connected and automated vehicles spans message standards, application platforms, cooperative-driving systems, radio measurements, and cellular trials. Each area provides an important part of the communication stack. However, our question crosses their usual boundaries: which complete CV and CAV exchanges can today’s direct and cellular paths support? We organize the literature around the two missing connections that this question exposes. The first is an application-interface gap between interoperable ITS messages and general stateful CAV services. The second is an evidence gap between path-specific measurements and a common, application-level readiness test.

2.1 V2X Standards and Application Platforms

SAE J2735 defines the content of widely used V2X messages, including Basic Safety Message (BSM), Personal Safety Message (PSM), Map Data (MAP), Signal Phase and Timing (SPaT), Signal Request Message (SRM), and Signal Status Message (SSM) [37]. Third Generation Partnership Project (3GPP) specifications complement these messages with network and application-layer support. In particular, TS 23.287 defines architectural support for New Radio (NR) V2X over

the PC5 and Uu interfaces [1]. The V2X Application Enabler (VAE) in TS 23.286 provides registration, group and message delivery, file distribution, network-resource management, service continuity, and session procedures. Its session-oriented service has identifiers and procedures for establishment, update, termination, notification, and acceptance or inability responses [4]. Similarly, the European Telecommunications Standards Institute (ETSI) Multi-access Edge Computing (MEC) V2X Information Service exposes information and subscriptions across vehicles, networks, and edge applications. It also includes predicted quality-of-service (QoS) information and V2X message publication and subscription. The message format can be attributed to a standards organization, and subscriptions can expire [17].

Broader architectures organize these functions. The 5G Automotive Association (5GAA) Application Layer Reference Architecture identifies application actors, deployment views, and reference points across the automotive ecosystem [5]. The ETSI Intelligent Transport Systems (ITS) station architecture defines a facilities layer with service-oriented application support. Its example functions include backend session establishment, session-loss handling, and session continuity during handover [16]. These architectures support interoperability and service deployment without prescribing one application data model.

Deployment platforms turn these standards into working applications. The United States Department of Transportation V2X Hub translates, aggregates, and disseminates messages between infrastructure components [43]. CARMA Platform integrates cooperative-driving applications with an automated vehicle stack [42]. Meanwhile, AutowareV2X integrates collective perception with Autoware and selects fresh messages delivered over two communication channels [8]. Thus, prior work already provides standardized content, application enablers, and deployed software.

Thus, prior work already supplies message content, session control, message services, and application-layer architectures. IPI addresses a narrower integration point. It defines a concrete, transport-independent application protocol that carries lightweight profiles for SAE J2735 message types or pre-encoded J2735 payloads together with correlated CAV service requests, updates, results, freshness bounds, and service states. This operation-level data model is chosen by IPI rather than by VAE, MEC, or the broader reference architectures. Therefore, IPI complements VAE session control and MEC service interfaces. It does not replace these standards or an automated-driving platform.

2.2 Networked CAV Applications and Middleware

Cooperative CAV systems show why this operation must cover more than compact safety messages. EMP sends vehicle point clouds to an edge server and merges the resulting perception across vehicles [48]. AutoCast instead selects useful lidar regions and schedules their direct transmission between vehicles [36]. Infrastructure-assisted systems make other representation choices. VIPS exchanges compact object descriptions for real-time perception fusion, VI-Map communicates representations for live high-definition map construction, and Soar coordinates several applications on deployed roadside infrastructure [22, 38, 39]. Together, these systems establish that raw sensor data, selected regions, features, detected objects, and maps impose different communication loads. They also show that representation is part of the application design.

Timeliness must likewise cover the complete application pipeline. D3 assigns dynamic deadlines to the operators of an autonomous-driving pipeline rather than treating each computation in isolation [21]. Tentacles extends this reasoning across communication and computation. It gives an application a common abstraction over multiple network interfaces. It then matches QoS requirements to its delay, bandwidth, reliability, and security characterizations. It can also combine paths, reduce data quality, or use a local result when a remote pipeline would miss its deadline [46]. Tentacles is therefore the closest middleware comparison to IPI.

However, these works solve a different interface problem. Each cooperative application defines its exchanged object and request lifecycle. Tentacles carries generic QoS, timing, and execution state, but it does not define a domain data model that spans SAE J2735 messages and general CAV service operations. Likewise, the SAE J2735 message dictionary alone does not define a general correlated service-result lifecycle. VAE session control and MEC message services provide complementary functions, while IPI defines the application operation and result representation used in this paper. A Tentacles-like system could then decide how to carry an IPI operation. Our evaluation places compact messages and larger CAV objects under the same complete-response criterion.

2.3 Direct V2X and PC5 Evaluation

Direct V2X studies have measured range, packet delivery, congestion, mobility, resource allocation, and channel behavior for several radio generations. Recent work also moves beyond packet-level safety beacons. SEE-V2X uses commercial Cellular V2X (C-V2X) radios and records application, medium access control, and sidelink-control traces for mixed BSM and perception traffic [30]. Ku et al. transfer fragmented

image frames over commercial Release 14 sidelink equipment. They report complete-frame loss and goodput while varying modulation and coding, packet size, location, and speed [27]. Separately, National Institute of Standards and Technology datasets examine standards interoperability among commercial on-board and roadside units [40]. Zhan et al. connect physical PC5 radios to a cooperative-driving controller in hardware-in-the-loop experiments [47]. CooperScene aligns commercial C-V2X measurements with synchronized multi-agent sensors and cooperative-perception benchmarks [45]. Together, these works provide commercial hardware, field or mobility context, application-shaped traffic, cross-layer traces, complete-frame outcomes, controller behavior, and interoperability evidence. In particular, SEE-V2X is a cross-layer traffic dataset, whereas Ku et al. directly evaluate complete image-frame delivery. Most closely, a recent 40-km private corridor evaluates both PC5 and Uu across six standards-based C-ITS scenarios; we discuss its broader matrix below [15].

The remaining evidence gap is therefore not a lack of realistic PC5 testing. Our study asks a different cross-application question. It places conventional-message functional checks and application-derived object classes under one all-attempt path-response criterion. PC5 performance uses a custom sequence-matched echo after separate BSM and SPaT functional checks, whereas Uu performance uses decoded, accepted IPI probe ACKs. The commonality is the application-level decision rule, not a shared wire protocol or a completed remote service. The direct component measures a production, standards-oriented off-the-shelf Long-Term Evolution C-V2X (LTE C-V2X) on-board-unit-roadside-unit path. It retains every attempted exchange in the availability denominator and reports round-trip time only for received responses. These results are specific to the measured LTE C-V2X equipment and conditions; they are not limits for NR-V2X or every PC5 implementation. The separately deployed 5G path applies the same decision rule to a vehicle-to-edge connection.

2.4 Cellular, 5G, and Edge Support for CAVs

Cellular studies examine vehicle applications at several levels. An initial commercial-5G study by Carpenter et al. measures delivery of camera, lidar, and other vehicle data and evaluates compression and encryption costs [9]. Kakkavas et al. evaluate a complete remote driving service on a real vehicle over a commercial fifth-generation (5G) Non-Standalone network. Their service sends video and sensor data uplink and control commands downlink to a center 36 km away [26]. A broader commercial-network study by Fezeu et al. connects physical-layer conditions, resource allocation, and handovers to camera and lidar streaming, per-frame delay and quality-of-experience (QoE), transport behavior, and teleoperation

quality [19]. Therefore, prior work already offers detailed public-network and application-specific evidence.

Private and experimental deployments answer complementary questions. TARGET-X reports automotive trials for cooperative perception, a digital twin, and predictive QoS for teleoperated driving [31]. A recent 40-km 5G Standalone private corridor is an especially close comparator. It evaluates PC5 and Uu across six Cooperative Intelligent Transport Systems (C-ITS) scenarios. It reports application-layer end-to-end latency and packet delivery while varying vehicle speed and traffic density [15]. In addition, the ongoing National Science Foundation DRIVE-SAFE project targets mobility, handover, tail latency, and integration of 5G with C-V2X for remote and cooperative driving [33]. We treat this project as evidence of an active research direction, not as a completed performance result.

These studies establish that our work is neither the first automotive 5G trial nor the first real-vehicle, uplink, private-network, or remote-operation study. Continuous teleoperation streams also appropriately use per-frame freshness and QoE measures. Our complete-response criterion instead applies to the stateless updates and correlated service operations represented by IPI. The evaluation adds one matrix across compact state, detector-sized objects, and larger transfers while varying payload, direction, transport, operating condition, offered uplink demand, logical-client concurrency, and service interruption. Unlike the corridor’s integrated NR-V2X deployment, our comparison uses separately deployed LTE C-V2X PC5 and local 5G NR Uu paths. The private 5G component reports availability across all attempts and round-trip time tails among received path responses. Because the endpoint clocks were not synchronized, it does not claim one-way latency.

2.5 Multi-Radio Communication and Co-Design

The two paths should ultimately cooperate rather than remain isolated. Multi-radio CAV architectures have long proposed assigning services to different radio access technologies [29]. AutowareV2X demonstrates redundant delivery and freshness-based selection, while Tentacles selects or combines interfaces according to application QoS and characterized network capability [8, 46]. More broadly, the 4C framework argues for joint design across computation, communication, and vehicle control [28]. These works establish the value of path diversity and application-network coordination.

Existing selectors can operate when one application provides its own state and freshness rules. However, a selector that generalizes across heterogeneous CAV services benefits from common operation semantics. It must distinguish

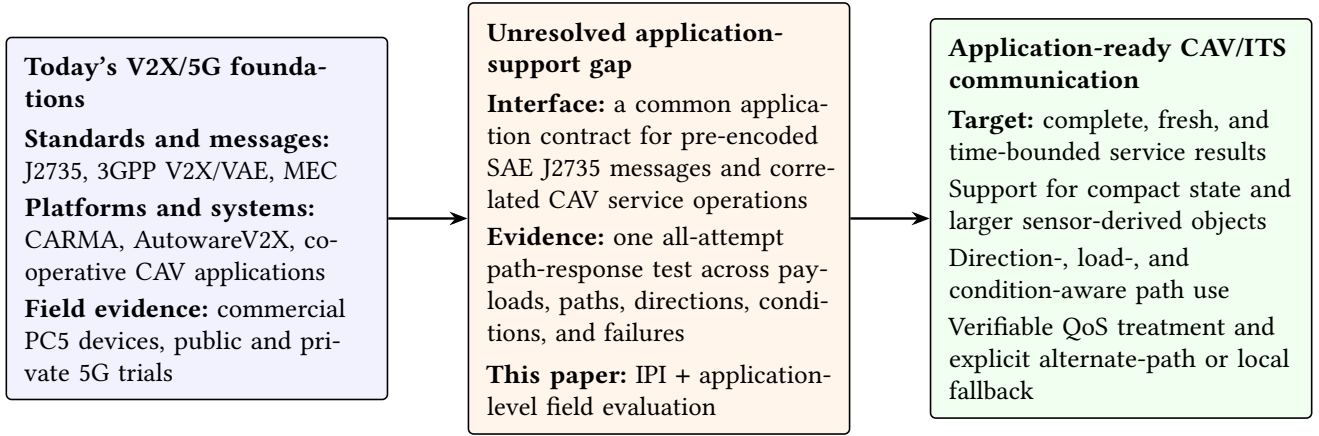


Figure 1: From today’s V2X/5G foundations to application-ready CAV/ITS communication. Existing technologies provide essential building blocks. IPI and the field evaluation address the two missing connections studied in this paper; they do not claim to implement every capability on the right.

which operation is in progress, which response completes it, and when the response expires. It also benefits from measured evidence about how path, payload, direction, signal, load, and service state change the outcome. IPI supplies the common representation, while our field evaluation supplies the measured inputs for the tested PC5 and Uu paths. This paper does not design a new path selector.

2.6 Synthesis

Taken together, prior work supplies standardized message content, V2X application and MEC services, cooperative applications, multi-network adaptation, and substantial field evidence. This paper contributes a specific combination across those areas. IPI brings lightweight profiles for SAE J2735 message types, pre-encoded J2735 payloads, and a correlated CAV service-operation data model into one operation-level protocol. The field evaluation then compares separately deployed LTE C-V2X PC5 and local 5G NR Uu paths across the paper’s payload, transport, operating-condition, load, concurrency, and service-interruption matrix. As Fig. 1 summarizes, these contributions address the interface and evidence gaps together. The next section presents IPI before the paper defines the measured paths and readiness conditions.

3 IPI Protocol Design

The related work shows that vehicular standards already define interoperable messages, while cellular and edge standards already define sessions and delivery services. However, a CAV application still needs to express one complete operation. It must identify the requester, bind every response to the right request, state whether work is continuing or finished,

and determine whether the result remains useful. The Intersection Programming Interface (IPI) defines this application contract. It carries conventional connected vehicle (CV) messages and stateful automated-vehicle services through one typed interface. Thus, IPI complements SAE J2735 and the Third Generation Partnership Project (3GPP) V2X Application Enabler; it does not replace either standard [4, 37]. More specifically, IPI places selected local J2735 profiles, opaque J2735 message operations, and CAV service metadata behind one reference facade. VAE or MEC services can carry these operations, while network-aware middleware can choose their path [4, 17, 46]. The contribution is this concrete application data model rather than a replacement session or multi-network framework. The reference adapter represents this service lifecycle, but it does not enforce every transition or generate terminal service outcomes.

3.1 Goals and Scope

IPI has four design goals. First, the same interface should serve a vehicle, roadside unit, traffic signal, intersection application, edge service, or cloud service. Second, the interface should preserve pre-encoded J2735 payloads while providing lightweight local profiles for selected fields. These profiles are a reference-model convenience rather than a replacement for a deployed J2735 schema and codec. Third, a longer CAV exchange should expose request identity, progress, completion or rejection, and time bounds. Finally, these application semantics should remain stable when a deployment changes its communication path.

These goals also bound the contribution. IPI is not a radio, routing protocol, scheduler, perception model, planner, controller, or edge platform. Moreover, a transport or priority

label does not reserve radio resources or create a quality-of-service (QoS) flow. Authentication, authorization, confidentiality, and replay protection also remain deployment responsibilities. Instead, IPI defines the information that an application and an evaluator need at the communication boundary. A network can then use that information for admission, scheduling, or path selection, but such actions require a separate mechanism. Similarly, the current types expose transport, priority, requested horizon, expiration, and session state. They do not yet provide one complete descriptor for direction, deadline, nominal rate, burst or stream behavior, and expected duration. Thus, IPI exposes part of an application’s communication intent, but it does not claim a network-enforced reservation.

Accordingly, this paper’s protocol contribution is the application data contract: its object types, metadata, service vocabulary, and interpretation of completion. It is not a claim that every transport adapter or enforcement mechanism is already deployed.

3.2 One Interface, Two Interaction Modes

Figure 2 shows the protocol structure. The first mode carries conventional Intelligent Transportation Systems (ITS) and CV messages. A producer can ingest or request broadcast of a typed message, while a consumer can retrieve messages by type and time. This mode supports independent state updates, such as a Basic Safety Message (BSM), Personal Safety Message (PSM), Map Data (MAP), or Signal Phase and Timing (SPaT) message. It also supports related J2735 exchanges, such as a Signal Request Message (SRM) followed by a Signal Status Message (SSM). However, an independent broadcast does not need a session merely because the interface can carry one.

The second mode carries a stateful service operation. A vehicle registers a profile, uses a session ID for heartbeat and telemetry calls, submits an envelope-based service request, and retrieves responses by session. The current vocabulary includes perception, planning, control, computation, high-definition map, lane-keeping, and unprotected-left-turn assistance. Envelope-based message, registration, and service-request calls can share source, place, time, transport, and correlation metadata. However, direct heartbeat and telemetry calls carry only their session-scoped fields. Only the stateful mode uses the service-lifecycle vocabulary.

3.3 Common Envelope and J2735 Message Path

Every message or service request submitted through the envelope interface carries common metadata. As Table 1 shows, the contract identifies the message identifier (ID), send time, intersection or service domain, selected transport, source

role, and source identifier. A session identifier scopes a continuing exchange. Likewise, a correlation identifier binds a response to a request, while an optional priority carries application intent. Session-scoped heartbeat and telemetry calls instead carry the session ID directly. The reference facade generates a local counter-based message ID and send time when the caller omits them. It also derives the correlation ID from a service request ID when needed. These are prototype policies, not a distributed uniqueness service. In particular, the default message IDs and derived correlation IDs are process-local convenience values. An application that requires uniqueness across vehicles, restarts, or concurrent requesters must supply its own correlation ID because a 16-bit request ID is not globally unique.

The message path then pairs this envelope with a payload descriptor. The descriptor records a message type, encoding, byte string, and optional frame counter. The typed reference path models BSM, PSM, MAP, SPaT, SRM, and SSM. Meanwhile, Traveler Information Message (TIM) and other messages can use an opaque path that preserves their pre-encoded bytes and declared type. Thus, an application can adopt IPI without waiting for every J2735 type to have a local C++ model.

For the selected reference profiles, the library provides project-local bit-packed and canonical encodings. Typed unpacking checks the requested application type and applies the validator implemented for that profile. However, these encodings are not a complete SAE J2735 implementation of Abstract Syntax Notation One (ASN.1) and its Unaligned Packed Encoding Rules (UPER). In particular, the current IPI-CooperativeService UPER and BYTES options both carry the same canonical IPI bytes. The custom cooperative object is a research-format proposal for a future deployment-specific J2735 extension path. It is neither an SAE-assigned message nor a validated ASN.1 regional extension. The device experiments instead use the vendor’s 2020 J2735 stack for selected standard messages on the PC5 path. Finally, the type vocabulary declares a JavaScript Object Notation (JSON) encoding, but the current codec rejects it as unimplemented. For the exercised profiles, parsers reject selected range, declared-length, truncation, and requested-type errors. However, they are not complete hostile-input or semantic validators. Some canonical parsers accept trailing bytes or unknown enumeration values, and a cooperative message need not match one service class to one content section.

3.4 Stateful CAV Service Contract

An `IpiServiceRequest` names the requested capability and assigns a 16-bit request identifier. It can also carry a desired

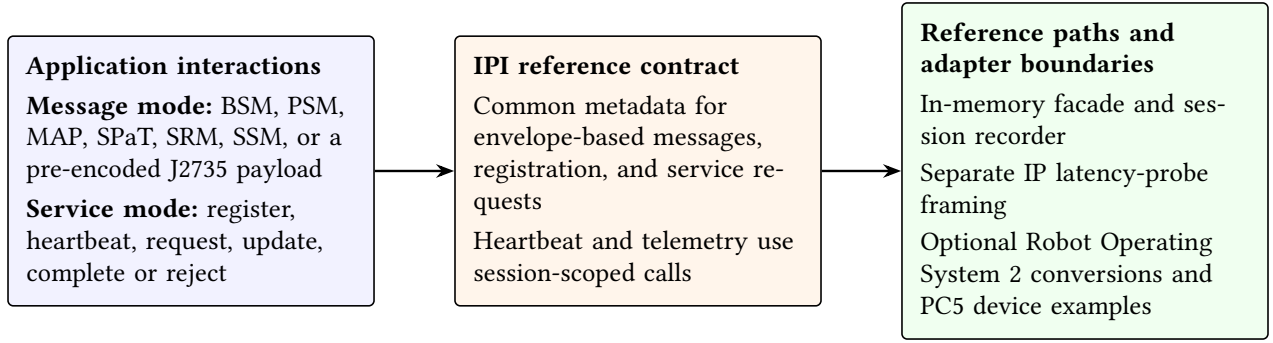


Figure 2: IPI separates an application operation from its communication path. The source tree contains a common in-memory facade, optional conversion and device examples, and separate field-measurement probe paths. The next section identifies which paths are exercised in the field.

horizon and opaque service context. The facade maps lane-keeping and unprotected-left-turn requests to either a CV or CAV contract according to the registered vehicle role. In contrast, perception, planning, control, computation, and map-update services require a CAV role in the current implementation. The envelope adds vehicle location and optional speed, heading, vehicle identification number, session, and correlation information. Therefore, the request can express who needs what service, where, and for which exchange.

The canonical request encoding uses big-endian multibyte fields. It places a one-byte service type before a two-byte request ID and one-byte presence flag. An optional two-byte horizon follows. Finally, a two-byte length prefixes the opaque context. This encoding is specific to the current reference version and limits both the horizon and context length as reported in Table 1.

Opaque context is important for extension, but it does not imply a standardized schema. For example, a disabled CAV can request planning assistance and place a fault report or remaining-capability description in the context field. An edge assistant can then return a path that moves the vehicle to a safe location. However, the current profile does not define a typed diagnostic or remote-recovery schema. This distinction permits the use case without overstating interoperability.

The corresponding cooperative-service object carries the evolving result. It uses a 16-byte session identifier, a 1–16-byte vehicle identifier, a service class, and one of four states: request, update, complete, or reject. Guided planning carries up to 50 waypoints, a fallback-route flag, and optional target speed or dwell time. Guided perception carries up to 64 objects with class, position, optional velocity, and optional covariance bytes. Guided control carries up to 10 bounded steering, throttle, or brake commands. The object can also carry a horizon of at most 60 s, confidence from 0 to 100, an expiration time, and optional computation-offload data.

These are limits of the current protocol profile, not limits of the network. Moreover, the control object is data rather than an authorization or actuation interface; a vehicle safety policy must decide whether and how to use it.

The cooperative encoding begins with the 16-byte session ID. It then carries a one-byte vehicle-ID length, the vehicle ID, a one-byte service class, a one-byte state, and a one-byte presence bitmap. Optional scalar fields follow the bitmap. Each planning, perception, control, or offload section is separately length-prefixed. Consequently, a receiver can skip an absent section and reject a section whose declared length exceeds the available bytes.

Together, the protocol types describe the intended lifecycle: *request* → *accepted or in progress* → *zero or more updates* → *complete or reject*. The application-facing response describes service handling, while the cooperative object describes the guidance exchange. Most importantly, a transport acknowledgment confirms only that an interface accepted an object. It does not mean that the requested work completed. The in-memory reference is narrower than the type model. Registration returns an active session, a known-session heartbeat restores the active state, and a session invocation records an in-progress response. It does not validate that an invocation or telemetry submission names an active session, generate completed or rejected service responses, enforce transition ordering, or advance a request automatically to completion. Therefore, the sequence is a protocol model rather than an enforced state machine in the current adapter.

3.5 Sessions, Bindings, and Failure Boundary

A registration accepts an envelope, vehicle profile, requested services, and optional event-subscription data. It can also name a roadside-unit (RSU) fallback and a minimum sidelink

Table 1: IPI field groups and current reference behavior. Required and optional identify the intended object schema; they do not imply that every listed field is enforced by the reference implementation.

Field group	Required	Optional	Semantic purpose	Current validation or behavior
Envelope	Message ID, send time, intersection, transport, source role and ID	Session ID, correlation ID, priority	Identifies an operation and its origin	Facade fills a missing ID and time; empty source and intersection IDs are not yet rejected
J2735 payload	Message type, encoding, bytes	Frame counter	Preserves a typed or pre-encoded vehicular message	Rejects empty opaque payloads; typed unpacking decodes a selected project profile, while opaque bytes are not decoded
Service request	Service type, request ID, vehicle ID, location	Vehicle identification number, speed, heading, horizon, context bytes	Names, correlates, and bounds requested work	Checks vehicle role, a 10-s horizon, and at most 65,535 context bytes
Cooperative result	Session ID, vehicle ID, service class, service state	Horizon, confidence, expiration, typed content, offload fields	Carries partial or terminal planning, perception, or control output	Checks selected ranges and payload counts; it does not enforce class/content consistency
Session	Session ID, vehicle profile, transport, state, lease, heartbeat period	Requested services, preferred channels, fallback roadside unit, sidelink-signal threshold	Scopes repeated requests, telemetry, and responses	Returns fixed lease and heartbeat values; fallback inputs are not retained
Response and acknowledgment	Service status or accepted flag	Expiration, guidance, de-tail ID	Separates interface acceptance from service completion	Adapter emits accepted or in-progress responses; terminal outcomes and structured failure codes are not generated

received-signal-strength indication (RSSI). The descriptor returns the session ID, vehicle profile, transport annotation, state, lease, heartbeat period, preferred channels, and granted services. The type system declares registered, active, suspended, and terminated states. However, the in-memory receiver returns active on registration and does not implement lease expiry or a complete transition machine. The high-level facade exposes registration, heartbeat, invocation, telemetry, and response retrieval. Patch and termination remain lower-level receiver operations.

The reference library defines deterministic topic names for registration, events, heartbeat, service request, service update, telemetry, and CV response. However, its only concrete

session transport records registration, heartbeat, service-request, and telemetry publications in memory. The measured Message Queuing Telemetry Transport (MQTT) probe uses separate latency request and acknowledgment topics. Thus, it does not implement the session-topic convention. An MQTT session adapter could map the named operations to broker topics, but that adapter is outside the current reference. The repository also contains optional Robot Operating System 2 (ROS 2) conversions and separate vendor-device examples for selected standard-message callbacks. These examples are distinct from the in-memory facade and do not establish an end-to-end facade-to-PC5 or facade-to-ROS 2 binding.

Transport independence has a limited and testable meaning here. The envelope records a coarse path class without

placing Transmission Control Protocol (TCP), MQTT, User Datagram Protocol (UDP), PC5, or 5G New Radio (NR) fields in the application object. TCP, MQTT, and UDP are instead separate probe choices. The selected path still determines reliability, ordering, congestion response, packet-size limits, security, and network QoS. Likewise, registration accepts fallback-RSU and minimum-RSSI inputs as future-selection hooks, but the in-memory receiver does not retain or act on them after registration. Thus, IPI provides a common object vocabulary for a deployment that changes paths; it does not implement path selection, fallback, or continuity in this paper.

Time and failure fields have the same boundary. Send time, requested horizon, expiration, correlation, and terminal state let an application perform its own test of whether a matching result arrived while it remained useful. The current implementation also rejects several malformed typed inputs, invalid service roles, and empty opaque payloads. Heartbeat, patch, and termination calls reject an unknown session, but service invocation and telemetry do not yet perform the same directory check. The types can represent explicit service rejection and session termination. However, the reference does not universally enforce expiration, automatic lease timeout, or a complete machine-readable taxonomy for a missing response, transport interruption, restart, and fallback. The evaluation consequently keeps a harness timeout separate from an application deadline. It also records an unanswered attempt as unavailable rather than silently converting it into a success.

3.6 Reference Implementation and Validation

The C++17 reference library has four layers. Its core defines service requests, cooperative results, canonical encodings, and selected validation. Its V2X layer adds lightweight J2735 profiles, an opaque-byte path, and optional ROS 2 conversions. Its application layer supplies the common facade, sender and receiver interfaces, sessions, telemetry, and response queries. Finally, its experiment adapters carry a versioned private 5G probe request and acknowledgment over TCP, MQTT, and UDP. They log the condition, transport, frame type, request metadata, timing, and parser acceptance. The acknowledgment indicates whether the receiver accepted and inspected the probe frame; it is not a completed planning, perception, control, or recovery result. Moreover, the current `serviceSuccess` field is a caller-supplied annotation combined with acknowledgment acceptance. It is not an independently observed application outcome.

Seven automated CTests validate selected internal flows. They exercise BSM and PSM facade round trips; one SPaT probe loopback; one cooperative-message round trip; opaque

TIM byte preservation; rejection of one truncated PSM; one service-request topic parse/format round trip; registration, heartbeat, invocation, telemetry, and session-keyed response retrieval; and TCP/MQTT probe loopback. All seven pass in the current artifact. However, they do not cover MAP, SRM, or SSM serialization, every topic family, a broker-backed session binding, correlation enforcement, negative session states, lease expiry, fallback, external J2735 interoperability, or service execution and completion. Thus, the tests establish selected internal field preservation and validation behavior. They do not establish formal SAE conformance, production security, automatic fallback, or deadline satisfaction.

IPI therefore gives compact vehicular messages and CAV service operations a common application boundary without claiming to solve the network beneath it. The next section defines the separately measured PC5 custom-echo and 5G IPI-ACK paths. It also defines the common necessary-condition test applied to them.

4 System Design and Experimental Setup

Section 3 defined the application object that the evaluation must deliver. This section now defines the paths, workloads, and decision rules used to determine whether that delivery remains useful to a vehicle. The evaluation uses a real instrumented autonomous vehicle and one road environment. However, the hardware is not presented as a new testbed. Instead, it supplies controlled and reproducible instances of the two access modes that current Cellular Vehicle-to-Everything (C-V2X) architecture places at the center of vehicular communication.

4.1 Why We Measure PC5 and Uu

The Third Generation Partnership Project (3GPP) V2X architecture lets a vehicle communicate over the PC5 reference point or through the Uu reference point to a cellular network [1]. PC5 supports direct local communication among vehicles, infrastructure, and other road users without carrying the application data through a mobile core. In contrast, Uu carries traffic through the radio access and core networks to an application endpoint. Thus, it exposes the complete vehicle-to-edge uplink, return downlink, and service path. These modes are complementary. Neither mode is restricted to one application class.

We evaluate one currently deployable implementation of each mode. The direct path uses a commercial Long-Term Evolution (LTE) C-V2X on-board unit (OBU) and roadside unit (RSU). The network-assisted path uses an on-site Fifth-Generation (5G) New Radio (NR) deployment, a provisioned subscriber identity module (SIM), a vehicle gateway, a radio access network, a mobile core, local routing, and an edge

endpoint. These choices connect directly to the systems surveyed in Section 2 while avoiding a comparison between laboratory radio emulators.

The two paths are measured independently. No request leaves over PC5 and returns over Uu. Likewise, the evaluation implements no path selector, aggregation, or automatic radio fallback. It also does not measure Dedicated Short-Range Communications (DSRC), ITS-G5, NR sidelink, millimeter-wave, satellite, roaming, or public-network handover. Therefore, the conclusions apply to the measured production LTE PC5 path and the measured private 5G NR Uu deployment. They do not define a universal limit for either technology.

The 5G deployment is realistic in component classes and controllable in load, but it is not a replica of a commercial public network. During the experiments, one vehicle gateway was the only user equipment (UE) on the private deployment. We introduce explicit competing traffic and 1–100 logical application clients behind that UE. These controls stress the common gateway, Uu path, core path, transport, and edge responder. However, they do not reproduce independent background UEs, operator admission and scheduling policies, handover, roaming, or geographic backhaul.

4.2 End-to-End Paths and Timing Boundary

Figure 3 shows the measured boundary. The vehicle carries an experiment computer, a Mocar OBU, a Cisco Meraki MG52-HW cellular gateway, and a NovAtel OEM7 Global Navigation Satellite System (GNSS) source. The OBU communicates directly with a fixed HUALI RSU over LTE C-V2X PC5. The experiment computer also reaches the MG52 over a 1 Gbit/s full-duplex Ethernet link. The MG52 then uses NR Uu to reach an outdoor Airspan AirSpeed 2900 n48 gNodeB. The supported component-level boundary continues through the Cisco private 5G functions and an on-site MX250 to a directly connected edge host. The deployment also contains one remote MX68CW. We show it as management context because the retained evidence does not establish that application packets traverse it [12–14].

For PC5, a round-trip time (RTT) contains sender processing around the vendor application programming interface, two traversals of the direct radio path, RSU processing, and request/reply correlation. For 5G, RTT contains the vehicle host, Ethernet, cellular gateway, NR scheduling and transmission, gNodeB, private-core forwarding, local routing, edge receiver or broker, and the return path. Neither measurement decomposes these aggregates into radio, scheduler, core, broker, or endpoint components.

The vehicle and edge clocks were unsynchronized in most 5G runs. One follow-up also recorded a large endpoint-clock offset. Therefore, we report same-host RTT for each request and acknowledgment (ACK). The harness computes it from

wall-clock timestamps taken immediately before transmission and after ACK receipt. This measure does not require edge-clock synchronization, but a local clock adjustment can affect it. Stored uplink and downlink timestamps remain diagnostic fields and are not treated as one-way latency. The timed path begins with an encoded probe object and ends when the harness receives an accepted ACK. It excludes sensing, detector execution, planning, vehicle actuation, and human response.

4.3 Field Geometry and Coverage Context

Measurements took place around fixed roadside infrastructure in one road environment with nearby buildings, stationary collection points, and repeated driving routes. We transform the retained GNSS coordinates into a local metric frame for publication and omit the exact deployment coordinates. The PC5 and 5G stationary points are not necessarily paired. Thus, a point from one path is not treated as a simultaneous observation of the other path. Collection dates and weather labels are also descriptive context rather than controlled propagation variables.

The 5G coverage survey contains 16 points. Reference Signal Received Power (RSRP) ranges from -121 to -91 dBm, with a mean of -106.88 dBm. Reference Signal Received Quality (RSRQ) ranges from -19 to -10 dB, with a mean of -11 dB. The reported signal-to-noise ratio (SNR) ranges from 3.5 to 27.5 dB, with a mean of 16.44 dB. The survey was collected with an iPhone and therefore characterizes that handset as a separate UE rather than the MG52 carrying the application traffic. These sparse samples and five historical run markers describe spatial context. Their interpolation is not a calibrated propagation model or a per-request channel trace. In particular, the retained application logs do not provide contemporaneous per-request Channel Quality Indicator (CQI), Modulation and Coding Scheme (MCS), Block Error Rate (BLER), Hybrid Automatic Repeat Request (HARQ), Radio Link Control (RLC), Physical Resource Block (PRB), scheduler, queue, or QoS Flow Identifier (QFI) state.

The direct device exposed no usable radio metric. Its signal diagnostic was empty. Therefore, we derive a route-level application-quality proxy from GNSS-joined successes, timeouts, and successful-response RTTs. We use it only as qualitative spatial context beside the route geometry. It is neither measured radio power nor an RF-quality ranking. One nearby point is non-line-of-sight (NLOS) because a building intersects its path to the RSU. Consequently, distance alone is not used as a signal-strength estimate.

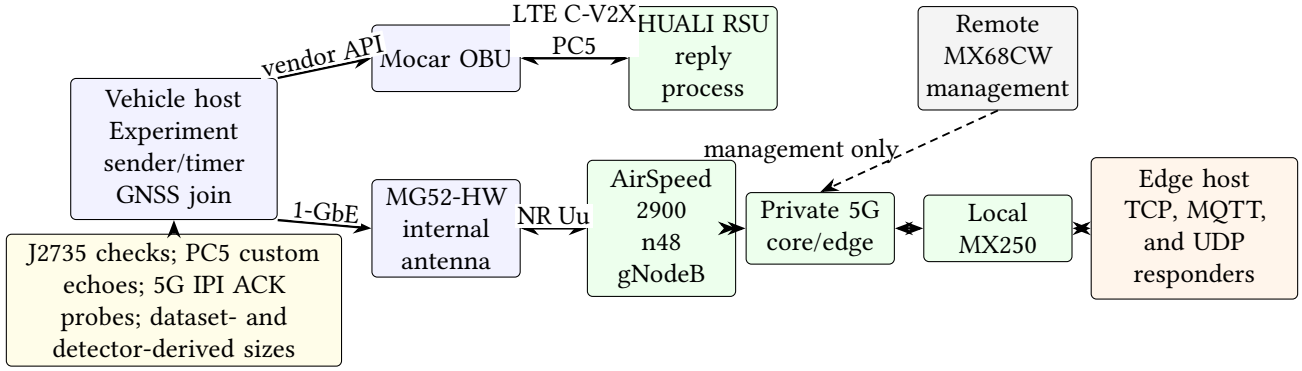


Figure 3: The two separately measured end-to-end paths. The PC5 path uses production OBU/RSU hardware. The Uu path contains one vehicle host, one MG52-HW/UE attachment, the n48 radio and private 5G functions, local routing, and an on-site edge endpoint. The remote MX68CW is not placed on the measured user-data path.

4.4 Direct LTE C-V2X PC5 Path

The vehicle-side Mocar OBU and fixed HUALI RSU use the vendor LTE C-V2X PC5 stack and 2020 J2735 software development kit. The vendor reports that these product families passed LTE-V2X protocol-conformance certification under the cited Chinese communication and certification rules [24]. This evidence supports their use as production, standards-oriented devices. It does not certify the performance measured in this paper.

Before measuring custom request/reply traffic, we validated the deployed standard-message path. An IPI bridge forwarded 10 of 10 encoded Signal Phase and Timing (SPaT) frames through the vendor SPaT interface to the OBU receive callback. We also observed Basic Safety Message (BSM) reception in both directions. These are functional checks rather than reliability estimates.

The installed public `mde_v2x_custom_send()` symbol did not transmit in this software build. Therefore, the probe uses the working `v2x_packet_data_send()` entry point with the experiment payload, length, and packet type `0x1b`. The probe adds a sequence number, request identifier, endpoint identifier, and echo response. Initial bring-up returned 10 of 10 correlated replies in both directions. The stationary and mobile sweeps then use the OBU as initiator and the fixed RSU as responder.

The resulting probe is not a standardized BSM, SPaT, or other J2735 frame. It is an addressed measurement message used after the production standard path was validated. Static analysis of the installed packet-data library identifies a 4,080-byte maximum application packet after vendor framing. The evaluated sweep stops at 2,048 bytes. Hence, an observed loss is not caused by submitting a payload above that interface limit.

The retained PC5 evidence includes device roles, message path, application limit, geometry, mobility, send interval, timeout, and application outcome. It does not include an exposed carrier, channel bandwidth, resource-pool mode, transmit power, antenna gain and placement, MCS policy, congestion-control configuration, firmware identifier, or lower-layer retransmission record. Accordingly, we report these fields as unavailable rather than inferring them from generic product capability. This limitation is materially different from the detailed exported configuration available for the 5G path.

4.5 Private 5G NR Uu Path

4.5.1 Radio access and spectrum. The outdoor access node is an Airspan AirSpeed 2900 (AS29-N48-DSC1). It integrates the central-unit control and user planes, distributed unit, and radio unit. The retained inventory records platform version 22.0-24-0.0 and application version 22.00-53-0.0. Its two enabled cells map one-to-one to two sector carriers. Both use NR band n48 in Citizens Broadband Radio Service (CBRS) spectrum, time-division duplexing (TDD), 40 MHz bandwidth, 30 kHz control and synchronization-signal sub-carrier spacing, and a 20 ms Synchronization Signal Block period. The active configuration permits at most one uplink layer and two downlink layers and enables downlink 256-quadrature-amplitude modulation as a cell capability. These are configuration limits, not the rank or modulation used for a particular request.

Table 2 records the operating context and its evidence boundary, including both cell channel records. Current evidence associates the original traffic with Cell 2, and the operator verified that the follow-up MG52 was locked to Cell 2; matched per-bin exports are not retained. Cell 1 is not a second application path. Each active CBRS export reports Category B General Authorized Access operation, one

requested and authorized grant, no suspended grant, and a successful Spectrum Access System heartbeat. The export also reports 33 dBm cell transmit power and 34 dBm/MHz cell Effective Isotropic Radiated Power (EIRP).

The integrated antenna has 17 dBi configured gain and cross polarization. The equipment guide describes a dual-slant pattern with about a 65° azimuth and 8° elevation beamwidth [7]. A signed structural record places its centerline 20 ft above roof level on a 21-ft tripod/mount structure. The installed height above ground, north-referenced azimuth, and mechanical and electrical downtilt are unresolved. Therefore, we do not replace the structural record with a propagation-model height or infer a measured beam footprint from the equipment pattern.

The original campaigns use 40/40/20. Its fields denote downlink, uplink, and dynamic frames. The later locked-cell campaign uses 70/20/10. It is downlink-heavy and reduces the uplink allocation. Both retain 10D4G, which the administrator identifies as frame packing for LTE coexistence. These labels are deployment shorthand, not self-defining 3GPP slot notation. An attempted 30/60/10 setting was not supported as a usable experiment configuration and produced no valid measurement. Moreover, the original and follow-up campaigns differ in date and radio context. Their comparison is descriptive unless a same-cell, matched 40/40/20 reference is collected.

The network was isolated from other users during the campaigns, and the operator reported no interference from other bands. Nevertheless, equal useful bytes can require different radio effort under different channel conditions because NR supports a wide range of modulation and coding efficiencies [2]. This motivates repeating load and concurrency tests at different field locations. Because matched PRB, MCS, BLER, HARQ, and scheduler counters are unavailable, the evaluation does not claim to measure the number of radio resources consumed by a weaker link.

4.5.2 Vehicle gateway, core, and edge. The vehicle gateway is a Cisco Meraki MG52-HW with an internal antenna and Telit FN990A40 modem. Cisco documents sub-6-GHz standalone and non-standalone 5G support and four downlink multiple-input multiple-output paths [11]. These are product capabilities rather than a measured rank. In every original and follow-up run, the gateway remains flat on the bottom of the trunk with its front face upward. This orientation matters because the internal antenna has a documented front-face bias [10]. The experiment Ethernet link negotiated 1,000 Mbit/s full duplex. Gateway, modem, host, and carrier software versions are MG 4.1.2, M0R.115005, A0R.501056, and P0R.000566, with the Generic GCF profile.

Provisioned SIMs attach the MG52 using Data Network Name (DNN) cisco5g. The operator identifies the session as

receiving default 5G QoS Identifier (5QI) 9 treatment. However, the numeric QFI and a verified nondefault QoS flow are unavailable. Likewise, one active export enables RLC acknowledged-mode capability, while another profile view displays bidirectional unacknowledged mode for 5QI profiles 1–9. Neither record proves the RLC mode of the measured bearer. We therefore do not attribute latency or loss to a particular QoS flow, RLC mode, or retransmission mechanism.

The edge host is directly connected to the on-site MX250. This removes an external cloud application hop after the local private 5G system. However, the exact User Plane Function (UPF), internal core route, and per-component delay are not retained. Thus, Figure 3 describes the path at the supported component boundary instead of assigning unmeasured latency to individual private-core functions.

4.6 Application Workloads and Protocol Conditions

The workload progresses from compact CV/ITS objects to a detector-derived size model and large application-object chunks. Standard BSM and SPaT traffic first validate the deployed PC5/J2735 path. Compact IPI conditions then use 0, 256, 1,024, and 4,096 bytes of application payload. These sizes represent control-only messages and compact state or service content; an arbitrary byte count is not presented as a named J2735 message.

Throughout this section, a named payload size denotes requested application or offload-payload bytes. The encoded IPI frame, probe metadata, and transport headers add bytes. The experiment logs record encoded IPI-frame bytes separately. We use named payload bytes to relate a condition to an application object class and report encoded sizes with the results.

For a CAV-derived communication workload, we run the official V2X-Radar radar-only late-fusion checkpoint over all 922 samples in the staged cooperative validation split. We estimate a detector-output payload as a 256-byte fixed header plus 96 bytes per predicted object. This is an application-level size model derived from prediction counts, not the detector's native serialization or a standardized perception-message encoding. The resulting minimum and median sizes are 13,216 and 19,648 bytes. The 95th, 99th, and maximum sizes are 22,816, 23,968, and 25,024 bytes. The network replay carries opaque byte payloads at representative sizes plus a separate 60,000-byte stress condition. The latter is not a detector percentile. Detector execution runs offline on a separate four-GPU host and is excluded from communication RTT.

The staged source manifest contains 22,431 artifacts from completed OpenDAIR-V2X, V2X-Seq, TruckV2X, and V2X-Radar sources. Encoded images span 25.5–666.9 KiB, with 334.1 KiB median, 539.8 KiB 95th percentile, and 648.9 KiB

Table 2: Private 5G operating context. A configured capability is not a measured per-request radio state.

Element	Value used or retained	Evidence	Interpretation boundary
RAN and spectrum	AirSpeed 2900; NR n48 TDD; 40 MHz; 30 kHz; 20-ms SSB	Active export	Configuration, not per-request MCS, rank, or scheduler state
Cells	Cell 1: ARFCN 637992, SSB 3558.720 MHz; Cell 2: ARFCN 645334, SSB 3669.600 MHz	Active export and operator verification	Records associate original traffic with Cell 2; the follow-up MG52 lock to Cell 2 was operator-verified; Cell 1 is not a second path
CBRS power/status	Category B GAA; 33 dBm transmit power; 34 dBm/MHz EIRP; authorized SAS heartbeat	CBSD export	Individual grant frequencies and per-request interference remain unavailable
Antenna/install	17 dBi, cross-polarized; product pattern about $65^\circ \times 8^\circ$; 20-ft centerline above roof level	Export, guide, structural record	Azimuth, downtilt, and installed height above ground are unresolved
TDD context	Original 40/40/20; follow-up 70/20/10; fixed 10D4G	Operator record	Deployment shorthand; historical comparison does not isolate causality
Vehicle gateway	MG52-HW, internal antenna, front face upward on trunk floor; 1-Gbit/s full-duplex Ethernet	Run record and link state	Fixed placement; no external-roof antenna or orientation comparison
Session/core	DNN cisco5g; operator-confirmed default 5QI 9; local MX250/edge host	Operator record and topology	QFI, bearer mode, UPF identity, and delay decomposition are unverified
Load/telemetry	One application UE; separate iPhone survey; host/application logs	Operator record and artifacts	iPhone is a separate UE; logical clients are not UEs; no matched per-request PRB/MCS/BLER data

99th percentile. Point clouds span about 15 KiB–3.85 MiB, with a 937.8 KiB median and 3.85 MiB 95th and 99th percentiles. The staged high-definition map files span 22.3–44.9 MiB, with 35.8 MiB median, 44.0 MiB 95th percentile, and 44.9 MiB 99th percentile. Accordingly, the 5G sweep adds 8, 16, 32, 64, 128, 256, and 512 KiB and 1 and 2 MiB conditions. These are transfer sizes or chunks. A completed chunk does not establish that a full point cloud or map met the same RTT.

Local IPI loopback checks construct, serialize, decode, and accept the selected payloads before wireless replay. They isolate format failure from network failure; their RTT is not a wireless baseline. Similarly, a 224-file four-GPU pass validates data staging and processing, while the 922-sample detector pass establishes the provenance of the detector sizes. Neither is a third communication path or a timed end-to-end driving loop.

The 5G experiments use three Internet Protocol (IP) conditions. Transmission Control Protocol (TCP) uses an addressed request/reply receiver. Message Queuing Telemetry Transport (MQTT) 3.1.1 uses QoS 0 over TCP with an edge broker and application receiver. Therefore, MQTT is an application-messaging condition over TCP, not an independent transport. User Datagram Protocol (UDP) is tested as 1) one raw datagram per logical object and 2) application-level fragmentation into datagrams capped at 1,400 bytes, followed by reassembly and an accepted ACK. The analysis never merges these two UDP variants.

The normal request carries the application object from vehicle to edge and a smaller ACK returns downlink. Separate 15-minute direction controls offer 25 Mbit/s in each direction. These controls measure achieved vehicle-to-edge and edge-to-vehicle throughput separately. They do not turn the request/reply RTT into one-way latency.

4.7 Application Readiness Criteria

No single latency, throughput, payload, or reliability value is “the CAV requirement.” A periodic signal message, a many-to-many warning, a per-vehicle trajectory, an emergency coordination burst, and a long-lived remote-assistance session place different demands on uplink, downlink, sidelink, edge state, and shared capacity. Therefore, we define readiness for each application in terms of the complete object, update pattern, direction, scope, duration, deadline, required probability, operating envelope, and failure behavior.

For a selected analysis budget B , the field instrumentation supports deadline-qualified path-response availability:

$$A_{\text{cycle}}(B) = \frac{N_{\text{path response, sender RTT} \leq B}}{N_{\text{sender attempts}}}. \quad (1)$$

A path response is a sequence-matched echo on PC5 or a decoded, accepted ACK on 5G. A timeout, missing response, rejected 5G frame, or response after B is a miss. RTT percentiles use only path responses and are always reported beside the all-attempt denominator. The 5G ACK confirms that the receiver decoded the IPI envelope and emitted an acknowledgment. It is not semantic validation of the opaque offload bytes, execution of a planning or recovery service, or enforced request/session correlation. For one-way periodic traffic, the correct quantity would instead be the fraction of updates whose age at the consumer is at most B . The current instrumentation does not measure that quantity. Hence, a full RTT is labeled an acknowledged-cycle comparison rather than one-way broadcast latency.

An application is supported only if its applicable conditions hold together: 1) the receiver interprets and correlates the complete object; 2) the object arrives within its useful time and required availability; 3) the path sustains the object size, rate, direction, and duration; 4) the condition remains true across the required geometry, radio state, load, and concurrency; and 5) failure is exposed soon enough for the application to stop, reduce, cache, prefetch, or choose another mode. Moreover, a deadline-critical exchange must retain its envelope while relevant periodic, bursty, bulk, and long-lived flows share the same resource. A favorable median among the replies that arrived cannot replace these joint conditions.

These are application-readiness conditions, not claims that every condition is directly measured by the present probes. In particular, the 5G probe measures IPI-envelope decoding and ACK return. It does not measure execution or semantic validation of an individualized planning or recovery result.

4.7.1 Representative CV and CAV envelopes. Table 3 selects five application classes before the results are examined. The early Vehicle Safety Communications (VSC) values are design targets, not current regulatory mandates [44]. The strict cooperative-driving and automated-intersection values come

from current 3GPP and 5G Automotive Association (5GAA) requirements [3, 6]. The experiment tests necessary communication conditions; it does not certify a safety application.

Vehicle motion makes these budgets concrete without creating them. At 25 mph (11.18 m/s), 100, 10, and 3 ms correspond to 1.12, 0.112, and 0.034 m of travel. Under the Federal Highway Administration roadway-design assumptions of 2.5 s perception/reaction time and 3.4 m/s² deceleration, the level-road stopping sight distance at 25 mph is about 46.3 m [18]. This is a human-warning reference, not an automated-braking model. For an illustrative blind-spot geometry with a 25-mph subject vehicle and a 35-mph adjacent vehicle, their same-direction gap changes by 0.45 m in 100 ms. These distances show the margin consumed by stale state; the cited application documents, not the distance calculation, define the communication targets.

The CAV examples also explain why direction and contention matter. Automated intersection crossing combines common downlink state with per-vehicle uplink intent and individualized downlink plans [23]. An emergency maneuver instead arrives as a short urgent burst. Finally, fault recovery first requires the vehicle to enter a minimal-risk state; an edge or remote assistant can then supply a high-level path toward a safer location [6, 32, 41]. At the cited assisted speed below 10 km/h, 100 and 200 ms correspond to at most 0.278 and 0.556 m of travel. The edge-local variant carries compact state and paths. The help-center variant is instead uplink-heavy because vehicle video dominates the returned path. Neither variant places network communication in the sole emergency-braking loop.

Figure 4 summarizes these distinct behaviors. It also shows why one peak-throughput number cannot establish support for all of them. Uu uplink and downlink occupy the configured TDD context but remain separate measured directions. PC5 uses a separate sidelink resource pool.

For cross-workload analysis, we report $A_{\text{cycle}}(B)$ at 100, 200, 500, and 1,000 ms. The 100-ms marker connects to the warning and some interactive examples. The 200-ms marker connects to high-level remote assistance. The 500- and 1,000-ms markers distinguish relaxed interactive service from asynchronous or bulk transfer; they are analytical thresholds, not universal standards. We also show 3-, 10-, and 25-ms markers for strict cooperative applications. Meeting a round-trip marker does not certify a one-way standard requirement. Likewise, 1,000 observed successes cannot validate a multi-nine reliability target.

4.8 Experiment Matrix and Analysis Rules

The experiments in Table 4 are derived from the readiness conditions rather than from the three findings previewed in the Introduction. They first establish functional message

Table 3: Representative traffic envelopes and the measurements that test a necessary condition. Payload values exclude protocol and security overhead.

Application	Traffic behavior and shared resource	Sourced object and time envelope	Necessary-condition test
Signal warning	Periodic common infrastructure-to-vehicle state; optional small request; PC5 pool or Uu downlink	About 500 B at 10 Hz; about 100 ms one-way reference	SPaT functional check; compact PC5 echo and 5G ACK screens; no one-way latency claim
Forward-collision or blind-spot warning	Periodic many-to-many PC5 awareness; a network realization creates recurring uplinks and warning fan-out	About 200 B per source at 10 Hz; about 100 ms and 150 m	256-B PC5 custom echo in stationary and mobile settings; compact 5G ACK screen
Automated intersection crossing	Recurring per-vehicle uplink state, common SPaT/MAP, individualized downlink plan, and edge session	300–450-B awareness at up to 10 Hz; 100-B SPaT; 1,000-B geometry; about 400-B plan; 10 ms and 99.9999%	SPaT functional check; 256/512-B PC5 custom echo; 1-KiB 5G IPI-envelope ACK; logical-client proxy. No MAP interoperability or plan execution is claimed
Emergency maneuver or trajectory alignment	Short event-driven reciprocal burst over PC5 or both Uu directions; must not wait behind bulk traffic	2,000 B at 3 or 10 ms; 12,000 B at 10 ms; 99.99–99.999% depending on use case	2-KiB PC5 custom echo and 8–16-KiB 5G ACK screens; no reproduced multi-vehicle maneuver or scheduler preemption
Fault-triggered remote recovery	Long-lived compact edge session, or sustained uplink video to a help center plus a small returned path	Edge: up to 1,000-B messages and 100-ms cycle; help center: about 32-Mbit/s video, up to 25-kbit/s path, 100/200-ms directional targets	1/4-KiB 5G ACK screens; direction control; 1-KiB foreground under uplink load; logical-client proxy and receiver restart

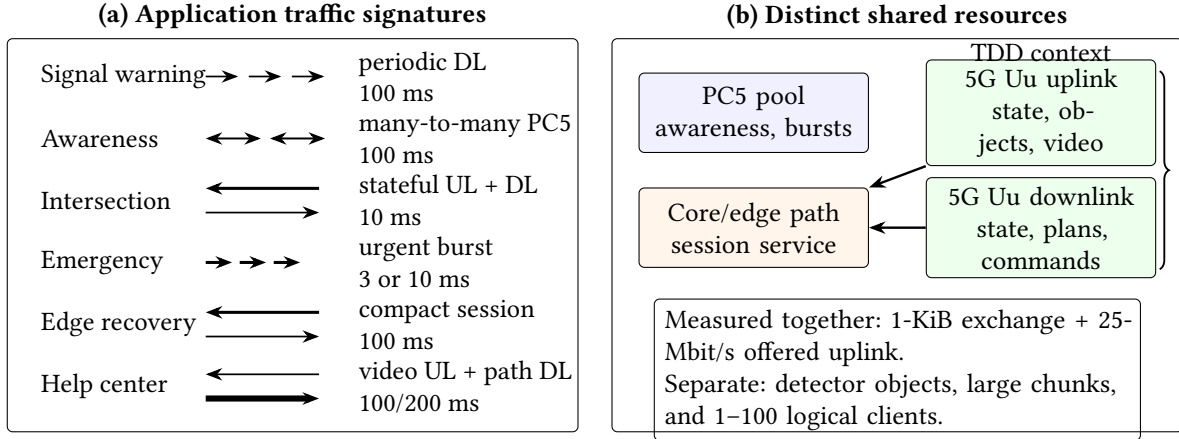


Figure 4: Application traffic and shared resources. Only the 1-KiB foreground exchange and offered uplink are measured concurrently; all other workloads run separately.

delivery. They then vary complete-object size, field condition, mobility, protocol, direction, offered uplink load, logical-client count, application deadline, QoS evidence, and service interruption.

The PC5 stationary sweep uses 0, 256, 512, 1,024, and 2,048 bytes, normally with 1,000 attempts per size, a 100-ms send interval, and a 1,000-ms timeout. Timeout-dominant conditions stopped by the operator retain their transmitted

Table 4: Experiment matrix. Most completed baseline conditions contain 1,000 attempts; exceptions and partial runs remain in the denominator.

Readiness question	Path and controlled factors	Fixed scope and repetitions	Outcomes
A. Functional and compact exchange	PC5: BSM/SPaT checks and echoes. 5G: 0–4-KiB IPI-envelope ACK probes over TCP/MQTT	Standard callback separate from each custom probe	Decode/acceptance, path-response availability, RTT
B. Object scale	PC5: 0–2 KiB. 5G: 0 B–2 MiB, detector-size model at 19,648–25,024 B, and 60,000-B stress	Usually 1,000 attempts; fixed per-run interval and sender timeout	$A_{\text{cycle}}(B)$ and p50/p95/p99 RTT
C. Coverage and mobility	PC5 stationary geometry and four mobile routes; 5G survey and stationary locations	Fixed hardware placement; GNSS join; no 5G handover trial	Spatial availability, path-response RTT, RSRP/RSRQ/SNR context
D. Direction and mixed-flow load	5G 1-KiB foreground idle or with offered 25-Mbit/s uplink; 1/2/4 streams; separate 15-min direction controls	One application UE; separate iPhone survey UE	Foreground $A_{\text{cycle}}(B)$, tails, and achieved rates
E. Logical-client demand	5G 1, 2, 5, 10, 20, 50, or 100 clients over TCP/MQTT/UDP	1-KiB sequential probes; per-maximum nominal 5 requests/s/client; one application UE	Aggregate and per-client ACK availability, rate, and RTT
F. Protocol and formation	5G TCP, MQTT QoS 0/TCP, raw UDP, and 1,400-B application-fragmented UDP	Complete message counted only after re-assembly and ACK	ACK availability, RTT, message-to-datagram amplification
G. Deadline, QoS evidence, interruption	5G 100/200/500/1,000-ms analysis; de-fault/5qi-mapped; receiver or broker restart	1-KiB request; 1,000 attempts per restart condition	$A_{\text{cycle}}(B)$, packet marking, misses, tails, recovery

attempts and identify unexecuted phases as skipped. The four mobile collections use 256-byte requests at 200-ms intervals. One stopped route contains 675 attempts; the other three contain 1,000. Their configured timeout is 500 or 1,000 ms.

The five 5G stationary payload campaigns were collected on May 13, 14, 15, 21, and 22, 2026. Most completed conditions contain 1,000 requests at 200-ms intervals. Detector replay uses TCP, MQTT, raw UDP, and fragmented UDP. The fragmented-UDP detector sizes normally use 1,000 attempts; the 60,000-byte stress condition uses 100. Depending on object size, its timeout is 3, 5, or 8 s. These harness timeouts define how long data collection waits. They are not application deadlines.

The heterogeneous mixed-flow experiment places a compact 1-KiB foreground exchange beside offered 25-Mbit/s vehicle-to-edge traffic. It reports the foreground deadline availability and the background’s achieved rate. Separately, the logical-client sweep creates 1, 2, 5, 10, 20, 50, or 100 clients. Each client uses sequential requests with a 200-ms post-attempt interval. Five requests/s/client, and hence 500

requests/s at 100 clients, is the maximum nominal rate when reply time is negligible. The realized rate can be lower while a client waits for a response or timeout. This is homogeneous application demand behind one UE, not a 100-vehicle field test. The locked-cell follow-up additionally uses four conditions: C1 is one 1-KiB client at idle, C2 is one 23,968-byte detector client at idle, C3 adds offered 25-Mbit/s uplink load to C1, and C4 uses 100 1-KiB clients. TCP, MQTT, and UDP occupy separate five-minute collection bins with two repetitions. C1–C3 use 500 probes per repetition; C4 uses 500 probes per client per repetition.

Finally, the QoS check captures vehicle-interface packets for default and 5qi-mapped application labels. This binary check asks whether the label changes observable host marking or establishes a bearer; it does not assume that it does. The interruption experiment restarts the TCP receiver, UDP receiver, MQTT application receiver, or MQTT broker while 1-KiB requests are active. It measures missed attempts and recovery. Because no alternate radio or local-computation

path is exercised, this is service interruption and recovery, not demonstrated fallback.

Across all families, path-response availability is the number of sender attempts that receive the path-specific response before the harness timeout divided by all sender attempts. The response is a sequence-matched PC5 echo or a decoded, accepted 5G ACK; the 5G sender does not independently enforce request/session correlation. Sender RTT is defined only for these responses. We report p50, p95, and p99 RTT with the denominator. We distinguish offered rate, achieved host throughput, application goodput when available, and request rate. Repeated requests are repeated observations of one field condition, not independent deployments. Failures, timeouts, partial files, and aborted diagnostics remain visible and are never replaced by retries in the reported denominator.

Together, these two paths cover the direct and network-assisted modes from Section 2. The experiment matrix moves from compact messages to detector-size objects and larger chunks, then varies spatial context, mobility, protocol, direction, uplink load, concurrency, deadline, QoS evidence, and service availability. Section 5 applies the same path-response and RTT definitions to determine where each measured path meets or misses the stated application envelope.

5 Experiment Results

Section 4 translated five connected-vehicle (CV) and connected and automated vehicle (CAV) applications into object, deadline, direction, and availability conditions. We now apply those conditions to the two measured paths. For every request/reply result, the denominator includes all sender attempts. We report round-trip time (RTT) percentiles only among accepted path responses and report $A_{\text{cycle}}(B)$ separately. For PC5, a path response is a sequence-matched echo. For 5G, it is a decoded, accepted acknowledgment (ACK) received within budget B ; the current sender does not independently verify request or session correlation. Thus, these results test communication necessary conditions. They do not measure planning correctness, vehicle actuation, or end-to-end safety.

5.1 Functional Message and Service Exchange

We first require the deployed receiver to decode the expected message and invoke the corresponding application path. Table 5 separates that binary check from performance evidence. The production Signal Phase and Timing (SPaT) bridge delivered 10 of 10 frames to the on-board-unit (OBU) callback, and Basic Safety Message (BSM) reception was observed in both directions. The local IPI validation also accepted compact controls and objects of 5,262, 5,789, 10,507, 40,337, and

60,000 B. These checks establish format and implementation coverage, not wireless reliability.

The direct custom bring-up returned 10 of 10 sequence-matched echoes. On the 5G path, all 2,000 selected SPaT/state-mirror probes received an accepted ACK. Their p50/p95/p99 RTTs were 58.748/130.323/140.479 ms. However, only 1,068 ACKs arrived within the 100-ms signal-warning reference, so $A_{\text{cycle}}(100 \text{ ms})$ was 53.40%. All 2,000 arrived within 500 and 1,000 ms. Because this is a request/ACK screen, the 53.40% value is not a one-way SPaT delay measurement.

Consequently, the conventional-message checks and the IPI envelope/ACK probes function at their stated format and path-response levels. These checks do not establish external interoperability, semantic service completion, or satisfaction of an application deadline.

5.2 PC5 Packet Size and Field Condition

The first PC5 readiness condition asks whether one complete local request/reply remains available as the requested payload grows and the field path changes. A 256-B probe approximates the roughly 200-B warning object in Table 3; 2,048 B approximates the cited 2,000-B cooperative-collision-avoidance object. The 10-, 25-, and 100-ms markers test strict cooperative and warning-scale timing. At 25 mph, 10 and 100 ms correspond to 0.112 and 1.12 m of travel, respectively. The measured quantity is still a two-way echo, not the one-way warning latency.

Figure 5(a) shows the all-attempt denominator. At the reference location, 256/512/1,024/2,048-B probes returned 999/1,000, 1,000/1,000, 1,000/1,000, and 1,000/1,000 replies. Their p50 RTTs were 99.980/99.972/99.966/99.958 ms, while p95 was 106.917/106.694/106.878/111.663 ms. Thus, even this favorable link placed the 10- and 25-ms lines below every median and the 100-ms line below every p95. As the field path weakened, packet size and radio condition interacted. The 2-KiB failure rate increased from 0.2% to 99.2%, followed by 910 observed failures before the next collection was stopped. Over the same sequence, the 1-KiB failure rate rose from 0% to 2.1% and then 82.9%. The 256- and 512-B conditions remained available longer, until the path became unavailable. The nearby non-line-of-sight (NLOS) point is shown separately because a building, not distance, blocks its path. Starred cells retain stopped or skipped phases in the analysis view while distinguishing them from transmitted failures.

Therefore, compact PC5 exchange is available under favorable local conditions, but the measured strict deadline and larger-packet conditions are not stable across the field paths.

Table 5: Functional checks and their claim boundaries.

Path	Pass condition	Outcome	Boundary
J2735	Standard callback	SPaT 10/10; BSM observed both ways	Functional sample only
IPI local	Decode and format-validate envelope/object	All listed sizes accepted	No wireless path or remote service execution
PC5 custom	Matched sequence echo	10/10	Bring-up only
5G state mirror	Decode and accepted ACK	2,000/2,000; 1,068 by 100 ms	No semantic result or one-way delay

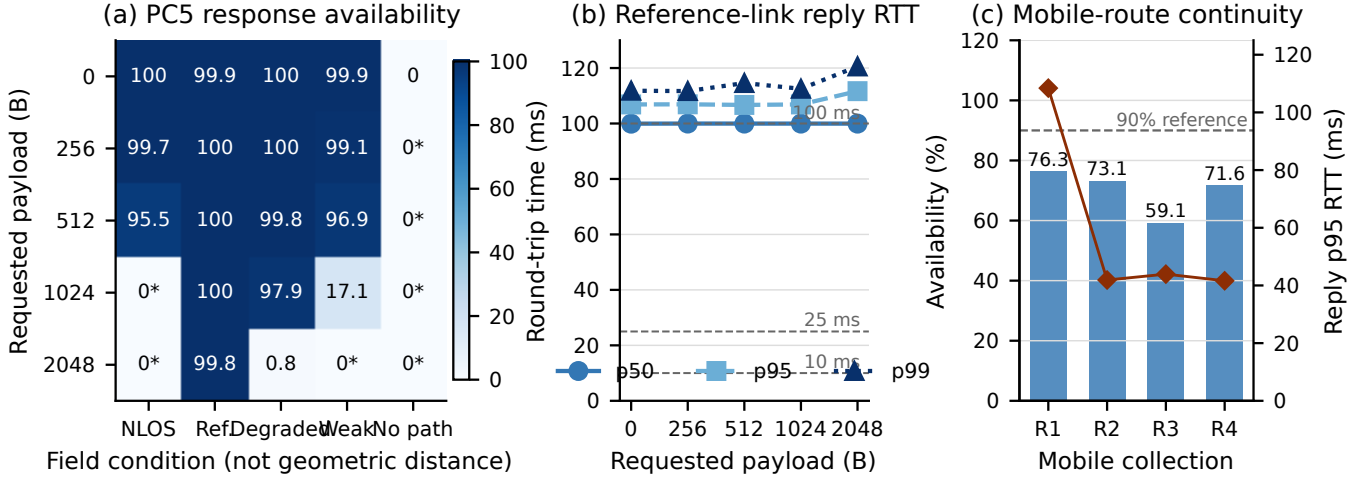


Figure 5: Direct LTE C-V2X PC5 comparisons. (a) All-attempt response availability for 0–2,048-B requests across distinct field conditions. Asterisks mark cells whose analysis view includes an operator-stopped or unexecuted phase; the prose reports observed failures separately. (b) Successful-response RTT at the stable reference location. (c) The four mobile collections returned 515/675, 731/1,000, 591/1,000, and 716/1,000 replies. The p95 line excludes missing requests. The 90% line is an illustrative lower reference from the application classes in Section 4, not a universal PC5 requirement.

5.3 PC5 Route Continuity

We next ask whether the same 256-B exchange remains available while the vehicle moves through the road environment. The four mobile collections achieved 76.3%, 73.1%, 59.1%, and 71.6% all-attempt response availability, as shown in Figure 5(c). Their accepted/attempt counts were 515/675, 731/1,000, 591/1,000, and 716/1,000. Among replies, p95 RTT was 108.421, 41.887, 43.923, and 41.621 ms, and p99 was 112.436, 55.880, 54.317, and 50.651 ms. Hence, the later route collections have sub-50-ms p95 among replies while 26.9–40.9% of requests receive no reply. The GNSS join places these misses in spatial weak- or no-path regions; the device exposed no usable received-signal-strength indicator, so we do not attribute them to mobility itself.

Accordingly, low latency where PC5 works does not compensate for route segments where the application receives no response.

5.4 5G Evidence Groups: Object Scale and Application Deadline

The 5G comparison asks which complete request/ACK objects arrive within 100, 500, or 1,000 ms. The object scale spans compact state, the 1–4-KiB fault and recovery-path bracket, the measured 23,968-B detector-output model, 128–512-KiB map/perception chunks, and transfers of at least 1 MiB. The 10- and 25-ms cooperative markers remain relevant, but no aggregate group in Figure 6 has a median accepted-response RTT below 25 ms.

The deadline-qualified availability changes substantially before eventual ACK availability does. Compact services at or below 4 KiB returned 60,000/60,000 ACKs, yet only 37.76% arrived within 100 ms; 96.29% arrived within 500 ms. The 34,000 loaded 1-KiB probes similarly achieved 24.12% at 100 ms and 93.47% at 500 ms. In contrast, 128–512-KiB objects returned all 18,000 ACKs, but only 0.99% arrived

within 100 ms and 88.36% within 500 ms. The 14,000 objects of at least 1 MiB also returned eventual ACKs, but none arrived within 500 ms and only 37.39% arrived within 1 s. These rows combine multiple conditions and are measured envelopes, not samples from one homogeneous deployment state.

Repeated campaigns show why the aggregate must not be read as one favorable link. In one stable collection, 0–4-KiB p95 was about 36–49 ms for Message Queuing Telemetry Transport (MQTT) and 100–140 ms for Transmission Control Protocol (TCP); at 2 MiB, p95 reached 2,536 and 3,633 ms, respectively. A second location produced a 10,767-ms TCP p95 at 2 MiB. Under a more variable path, even 0-B TCP and MQTT p95 rose to 1,142 and 510 ms, while the 256-KiB MQTT condition returned 999/1,000 ACKs with a 43,836-ms p95. These are collection-context differences, not weather-causality results.

Thus, acknowledged-cycle availability contracts under tighter budgets and for some larger objects across the evaluated combinations. This evidence does not form a monotonic payload-only curve or certify an application.

5.5 Protocol and Complete-Object Formation

We then compare how a zero-byte control, detector-derived object sizes, and a 60,000-B stress object are formed and transported. A pass requires one complete application object and a returned ACK. Therefore, receiving some Internet Protocol (IP) fragments is not a success. At 10 objects/s, the measured 23,968-B model would offer 1.917 Mbit/s of payload and 180 application datagrams/s at the tested 1,400-B fragment cap; the 60,000-B stress object would offer 4.8 Mbit/s and 430 datagrams/s. The experiment sends sequential probes, so this translation does not claim sustained 10-Hz fulfillment.

In the favorable-path collection, TCP and MQTT returned every ACK in each listed detector-size condition. For MQTT, 0-B, 23–24-KiB, and 60-KiB objects had p95 RTTs of 49.826, 183.705, and 403.790 ms; the corresponding TCP p95 values were 151.696, 169.540–213.894, and 381.409 ms. Under the weak condition, both transports still returned all selected objects, but 60-KiB MQTT and TCP p95/p99 rose to 770/1,230 and 662/1,027 ms.

The User Datagram Protocol (UDP) result differs. On the finalized raw-datagram path, 0 and 1,024 B returned 1,000 of 1,000 and 300 of 300 replies. In contrast, 1,400, 4,096, and 19,648 B returned 0/45, 0/100, and 0/100. After application fragmentation and receiver reassembly, 0–23,968-B objects returned 99.5–100.0% complete replies in the favorable-path collection, but the 60-KiB stress object returned only 20/100. Under the weak condition, objects through 4,096 B returned 99.9–100.0%; 19–25-KiB objects returned 93.5–96.1%; and

60 KiB returned 0/100. The earlier aborted UDP diagnostic is excluded.

Therefore, application fragmentation moves the measured UDP failure point, but it does not make large objects dependable in the weak-path collection; TCP and MQTT keep eventual delivery by accepting larger deadline-sensitive tails.

5.6 Direction, Competing Uplink Traffic, and QoS Evidence

The next comparison places the asymmetric application portfolio from Table 3 on the measured 5G path. The 15-minute vehicle-to-edge TCP control transferred 1,475,758,800 B at 13.118 Mbit/s. TCP backpressure kept it below the configured 25-Mbit/s offered target. The edge-to-vehicle control transferred 2,812,500,000 B at the configured 25 Mbit/s. Consequently, this vehicle-to-edge result does not demonstrate the 32-Mbit/s video class of the help-center recovery example. Conversely, the downlink rate exceeds its 25–400-kbit/s path/command class, but throughput alone does not establish the 100/200-ms or reliability requirements. These host-level controls do not identify a scheduler, Physical Resource Block (PRB), Modulation and Coding Scheme (MCS), or Time Division Duplex (TDD) cause.

These are host-side direction controls. The MG52-to-Cell 2 lock is operator-verified, and the 70/20/10 allocation is recorded in the experiment configuration. Without matched Airspan counters, these measurements do not isolate a TDD effect.

We then compare a 1-KiB foreground request/ACK at idle and while generated vehicle-originated traffic shares the same host, user-equipment (UE) attachment, Uu path, core path, and edge endpoint. The background generator targeted 25 Mbit/s per stream, but the two favorable repeats achieved only 1.320–2.267 Mbit/s at the vehicle host. Therefore, these runs compare the foreground while a best-effort generator is active; they do not establish coexistence under a sustained 25-Mbit/s load. In the favorable-path collection, the active generator increased MQTT p95 from about 52–60 to 174–176 ms and TCP p95 from about 162–194 to 388–420 ms. Across the two favorable repeats, MQTT A_{cycle} (100 ms) fell from 99.70% at idle to 40.15% under load, while its 200-ms value changed from 99.95% to 98.00%. TCP’s corresponding 100-ms values were 0.75% and 0.40%, while its 200-ms value fell from 97.90% to 59.00%. The one-, two-, and four-stream sweep achieved 1.990–5.743 Mbit/s. Under the default label, MQTT p95 was 170, 468, and 182 ms, while TCP p95 was 391, 402, and 410 ms. The non-monotonic MQTT sequence is retained because these were sequential field collections. In the first location-associated weak-path collection, completed client summaries achieved 0–0.322 Mbit/s across both application labels. The second collection has no comparable

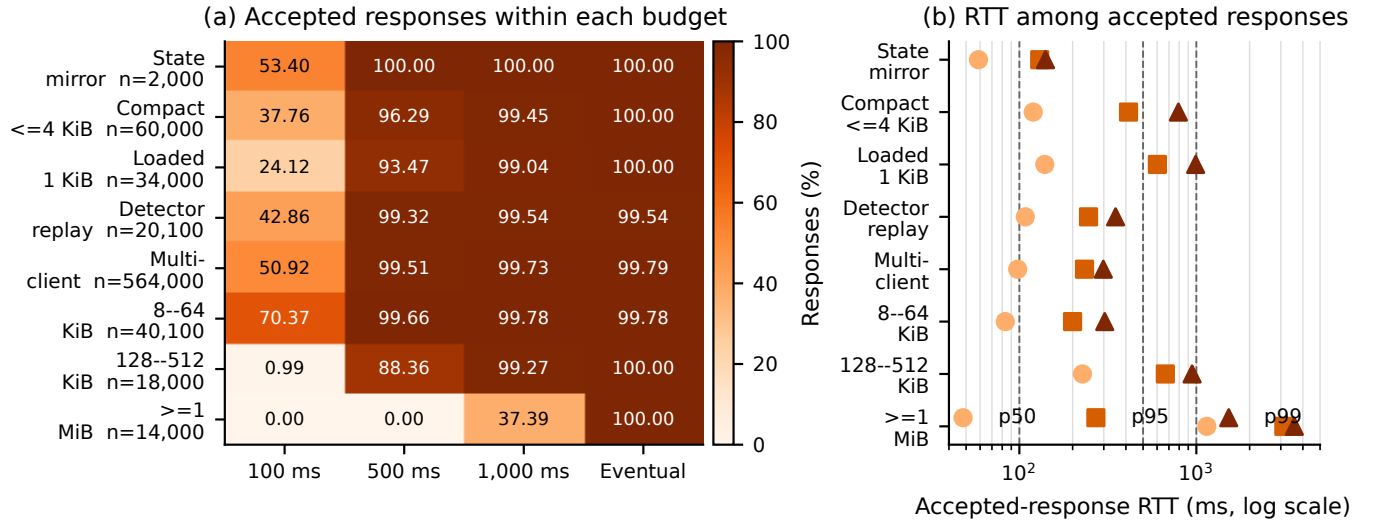


Figure 6: Private 5G Uu deadline comparison. (a) Every cell uses all sender attempts and reports $A_{\text{cycle}}(B)$, where a successful 5G cycle is a decoded, accepted ACK rather than a semantically completed CAV service. (b) RTT percentiles use only accepted responses and must be read beside the left-panel denominator. Each row aggregates the named evidence group across its collected field and load conditions. Thus, the rows are heterogeneous evidence groups, not a controlled payload sweep or a condition-matched performance curve.

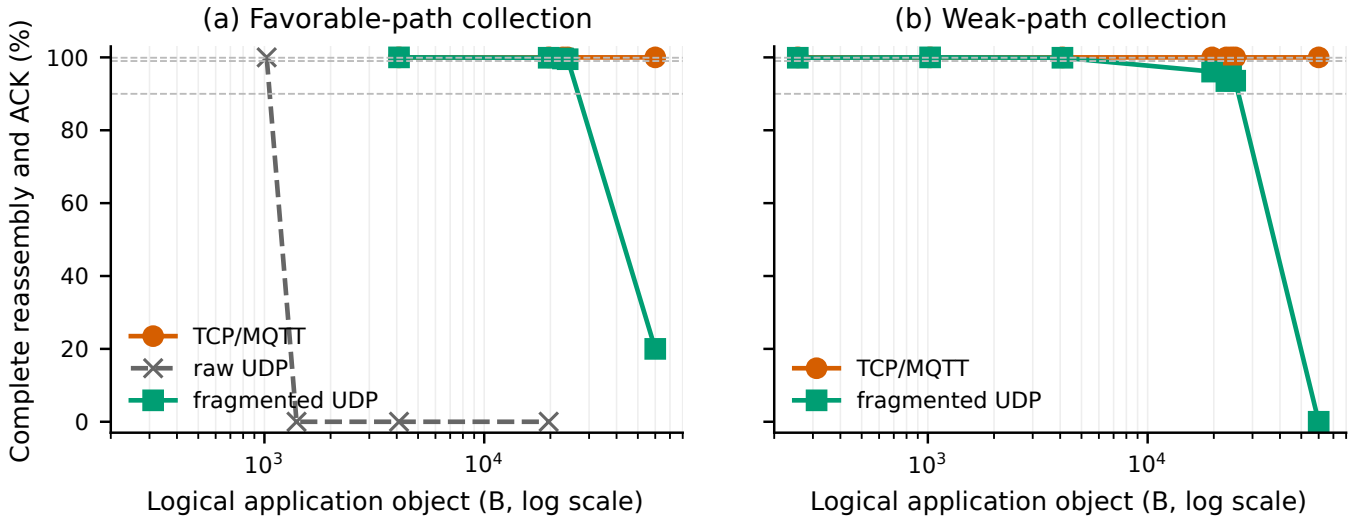


Figure 7: Complete-object response availability by protocol. TCP and MQTT are combined only where both returned all tested objects; their distinct RTT tails are reported in the text. Raw UDP sends one application object as one datagram. Fragmented UDP uses 1,400-B application fragments and counts a response only after receiver-side reassembly and ACK return. The horizontal lines are descriptive comparison levels, not universal requirements.

aggregate-rate summary. With the generator active, default-label p95 remained 460–770 ms for MQTT and 630–934 ms for TCP. Across the pooled active-generator streams, the 200-ms values were 83.63%/38.40% for MQTT/TCP in the first collection and 91.90%/49.17% in the second. The locked-cell

follow-up also retained five UDP ACK timeouts in its all-attempt denominator. Because each timed exchange includes both directions, these changes cannot be assigned to uplink scheduling alone.

Finally, the application-side 5qi-mapped label did not pass the predeclared Quality of Service (QoS) evidence check. The

default and labeled vehicle-interface captures both decoded observed packets with Type of Service (TOS) 0x0. Therefore, the different sequential p95 values under that label cannot be attributed to a verified nondefault 5G QoS Identifier (5QI), QoS Flow Identifier (QFI), or protected bearer.

Hence, the deployed path does not demonstrate the help-center uplink class, and a compact foreground develops larger tails while a best-effort background generator is active, without evidence of protected treatment.

5.7 Logical-Client Demand and Field Condition

We next scale one homogeneous 1-KiB request pattern from 1 to 100 logical clients. All clients share one host and one UE, so this is not a 100-vehicle or multi-UE scheduler experiment. Each client issues its next request only after the preceding request completes or times out and the configured interval elapses. Thus, five requests/s/client and 500 requests/s at 100 clients are maxima, not sustained offered rates. In the favorable-path collection, the complete seven-point repeat reached 100 clients with aggregate p95 RTTs of 81.850 ms for MQTT, 264.070 ms for TCP, and 113.391 ms for UDP. MQTT and TCP accepted 100,000/100,000 attempts, while UDP accepted 99,965/100,000. A separate 100-client repeat produced p95 values of 177.053, 283.765, and 145.894 ms and accepted 100,000, 100,000, and 99,365 attempts, respectively. The two repeats therefore remain distinct rather than selecting the smaller tail.

Under the location-associated weak-path collection, the 100-client aggregate p95 rose to 5,135.984 ms for MQTT and 2,908.420 ms for TCP. Their eventual response availability remained 99.499% and 100%. Across the 100 client-specific distributions, the median p95/p99 was 3,814.102/13,149.421 ms for MQTT. It was 2,938.408/6,400.146 ms for TCP. In contrast, UDP retained a 172.500-ms aggregate p95 but accepted only 90,811/100,000 attempts. Thus, comparing only the latency of replies would incorrectly favor UDP. The result establishes an application-level interaction between field condition and aggregate demand; it does not identify PRB allocation, retransmission, or scheduler behavior.

The multi-location follow-up is consistent with that scope. Its medium/typical block accepted 307,662 of 309,000 responses. The weak candidate, common repeat, and strong candidate C1/C2 blocks accepted 5,994, 5,999, and 6,000 of 6,000 responses. Using the two common blocks as a descriptive reference, the strong candidate had lower p95 in every matched C1/C2 transport aggregate. However, the blocks were sequential, and the iPhone radio observations characterize a separate UE rather than the MG52 application path. The association therefore cannot be assigned to Reference Signal Received Power (RSRP) alone.

Therefore, favorable-path capacity for compact requests does not predict the same deadline availability under weak-path concurrency.

5.8 Service Interruption and Recovery

The final necessary-condition test restarts a receiver or broker while 1-KiB requests are active. MQTT broker restart returned 955/1,000 responses, MQTT receiver restart returned 991/1,000, TCP receiver restart returned 953/1,000, and UDP receiver restart returned 969/1,000. Their p99 RTTs among replies were 90.553, 259.906, 361.228, and 63.027 ms. Nevertheless, the gaps between accepted replies were 15.1007, 13.3674, 11.7147, and 13.4635 s for a configured 10-s outage. At the 200-ms remote-assistance marker, A_{cycle} was 94.7%, 97.7%, 92.9%, and 96.9%, respectively. Thus, the completed-reply percentiles omit 9–47 missing attempts and the recovery interval. None of the four conditions reached 99.9% availability during the injected event. Moreover, no alternate radio, cached result, local computation, or safe-control action ran in this experiment.

Consequently, ordinary accepted-response percentiles can hide an application service gap that must be exposed before a faulted vehicle begins another assisted segment.

5.9 Summary, Discussion, and Insights

Together, the results give a conditional answer to the paper’s question. Today’s measured paths support selected communication functions and necessary communication conditions for CV and CAV applications. However, readiness depends on the application’s object, deadline, direction, traffic pattern, operating condition, and failure behavior. PC5 carries conventional messages and compact local exchanges under favorable conditions, yet larger packets and route gaps reduce availability. The 5G path carries compact and some detector-sized objects at relaxed deadlines, but the measured envelope contracts with tight budgets, larger objects, location-associated weak-path collections, active background traffic, concurrency, and service restart. Thus, neither a low successful-packet median nor eventual ACK delivery establishes application readiness. These conclusions apply to the tested devices and paths; they do not reproduce independent public-network UEs, handover, roaming, geographic backhaul, or operator scheduling policy.

The measurements lead to three concise insights. In these statements, “support” means meeting a measured necessary communication condition; it does not mean semantic service execution.

- **Current direct V2X paths cannot, by themselves, support complex CAV applications that require dependable large-object or multi-step service exchange.** On the measured PC5 path, larger packets,

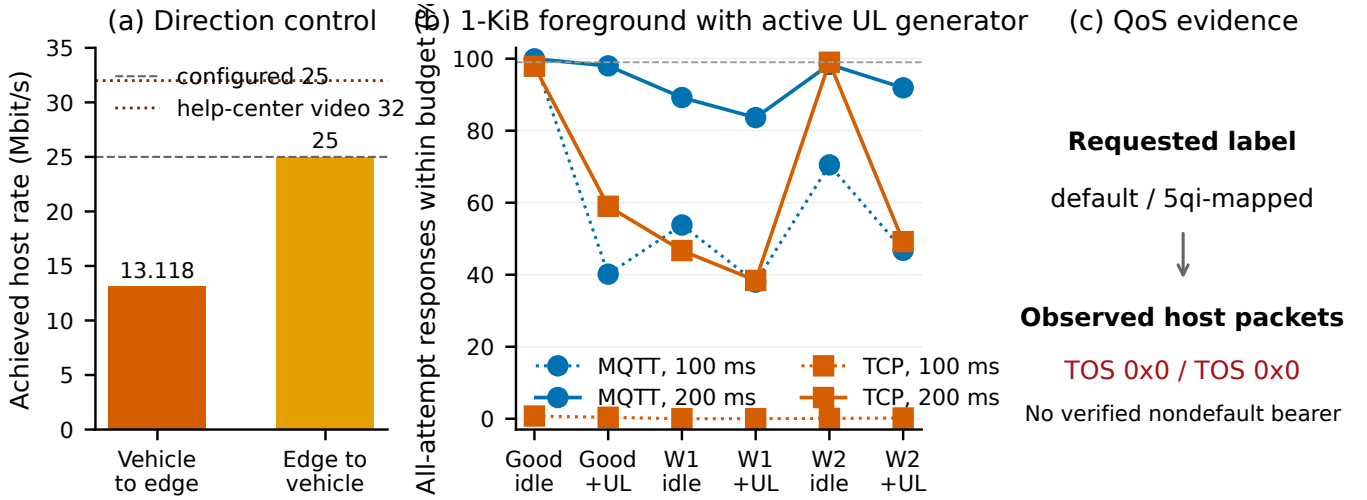


Figure 8: Direction, coexistence, and QoS evidence. (a) Achieved host-level direction controls beside the configured 25-Mbit/s line and the 32-Mbit/s help-center-video requirement. (b) All-attempt 100- and 200-ms availability of the compact foreground while a vehicle-originated generator is active. “W1” and “W2” are separate weak-path collections, and “+UL” pools their one-, two-, and four-stream default-label conditions. The favorable “+UL” repeats achieved 1.320–2.267 Mbit/s. W1’s completed default-label client summaries achieved 0–0.262 Mbit/s, while W2 has no comparable aggregate-rate summary; “+UL” therefore denotes an active best-effort generator rather than sustained 25-Mbit/s traffic. (c) Both application labels produced TOS 0x0 in the vehicle capture, so the intended nondefault treatment is unverified.

obstruction, and route gaps reduced response availability even when received replies had low RTT.

- **Today’s 5G can support some complex CAV exchanges, but the usable data size is strictly limited by the application’s time budget and operating conditions.** On the measured Uu path, the ACK envelope contracted with tighter deadlines, larger objects, location-associated path changes, active background traffic, concurrency, and interruption.
- **Collaborative CAV applications need communication methods beyond current V2X and 5G paths used independently and as-is.** The measured PC5 and Uu envelopes are complementary but incomplete; IPI defines common application semantics, while future systems must combine those semantics with verified representation, path, scheduling, admission, and fallback.

These insights turn the measured gaps into research questions. Accordingly, the next section identifies directions for the CAV, ITS, radio, edge, and carrier communities rather than proposing minor extensions to this specific deployment.

6 Future Research Directions

The results point beyond choosing between PC5 and 5G or increasing one peak data rate. Future CAV and Intelligent Transportation System (ITS) communication must make application requirements visible, protect different services when they coexist, adapt data and computation to the available path, preserve state through disruption, and evaluate these capabilities with application-level evidence. These needs define a research agenda for CAV, ITS, traffic-signal, radio, carrier, and edge communities. They are not additional insights or planned extensions to this deployment.

6.1 Turn Application Requirements into an Enforceable Network Service

The open question is how an application can state what makes one result usable and how the network can expose and enforce what it provides. An application knows whether a result is common or individualized, periodic or event-driven, deadline-critical or prefetchable, and useful in partial form. Conversely, the network observes radio condition, load, direction-specific capacity, mobility, bearer treatment, and predicted interruption. Existing V2X requirements already show that services have different direction, rate, latency, reliability, and recovery needs [3, 6]. However, a priority field,

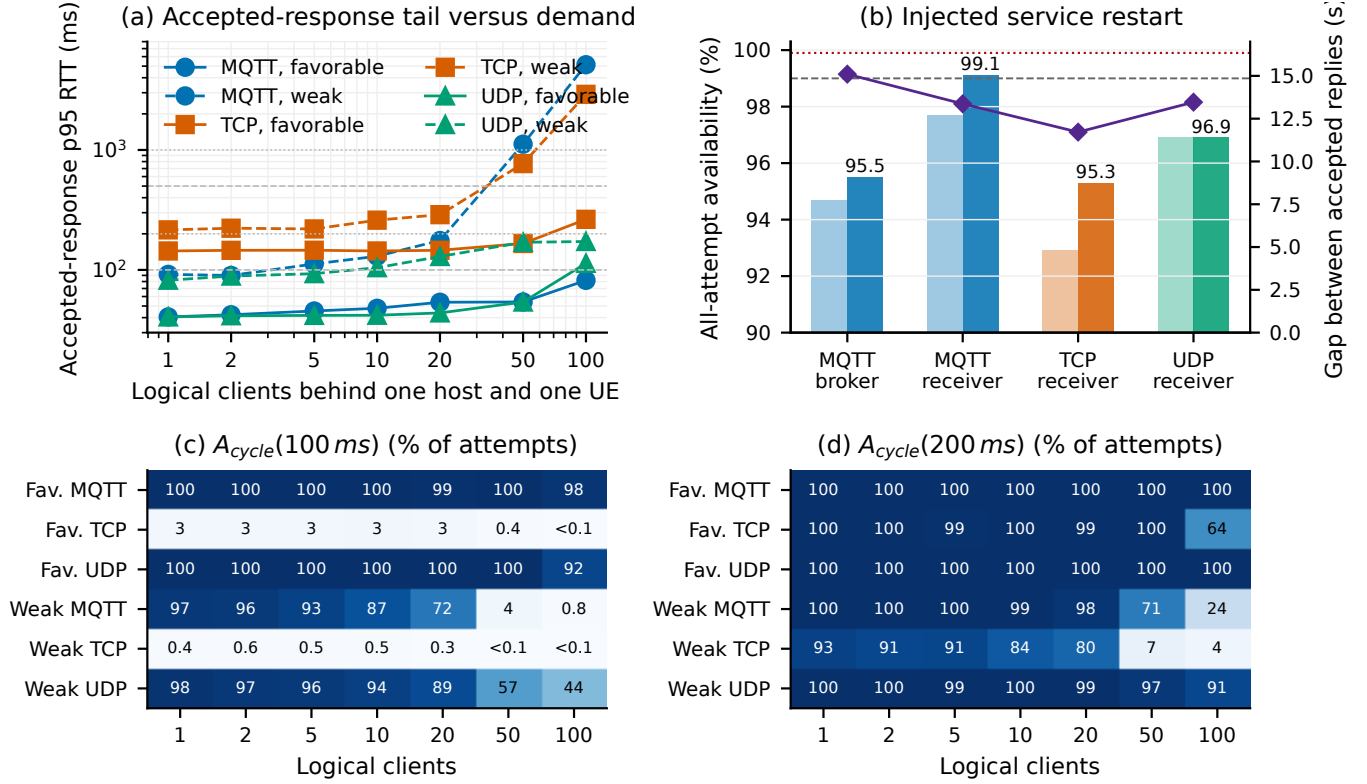


Figure 9: Demand and interruption expose failures hidden by successful-reply latency. (a) Aggregate p95 versus logical clients for the complete favorable repeat and weak-path repeat. Every client shares one host and UE. (b) During a 10-s injected restart, paired bars report $A_{cycle}(200\text{ ms})$ and eventual path-response availability over every attempt; the pale left bar is the 200-ms value and the dark right bar is eventual. The line shows the longer gap between accepted replies. The 99% and 99.9% lines are descriptive; 1,000 attempts cannot validate a multi-nine target. (c–d) All-attempt 100- and 200-ms availability is recomputed at every client count from the retained sender records. Values are percentages.

QoS name, or application programming interface option is not a service contract unless it maps to observable treatment.

Therefore, future work should define a machine-readable contract between applications and networks. The application side should describe one complete result: object size, update pattern, direction, delivery scope, deadline, freshness, required availability, acceptable partial result, and fallback action. The network side should return the resources or policy it admitted, what it can currently or predictively provide, and whether that treatment remains in force. This contract must support explicit rejection, negotiated degradation, revocation, and auditing. It must also authenticate declarations and limit inflated priority or rate requests. Progress should be measured by whether the declared requirement, admitted treatment, observed result, and failure state agree end to end. IPI provides part of the application vocabulary, while this broader outcome requires coordination across standards,

operating systems, radio access networks, edge platforms, and carrier policy.

6.2 Schedule Heterogeneous Vehicular Services Across Radio and Edge Resources

The open question is how radio, transport, and edge resources should admit and schedule mixed CAV services without starving other traffic. Future roads will mix periodic awareness, common signal state, individualized trajectories, urgent maneuver bursts, processed perception, bulk updates, and long-lived recovery streams. These services compete differently. PC5 uses sidelink resources, whereas Uu has direction-specific radio opportunities. Stateful services also occupy gateway, transport, core, edge-compute, backhaul, and sometimes remote-assistant capacity. Moreover, uplink-heavy verticals must operate within Time Division Duplex

(TDD), interference, and deployment constraints rather than assuming symmetric capacity [34].

Accordingly, schedulers and admission control should optimize complete useful results, not isolated packet delay or aggregate throughput. Research is needed on deadline-aware admission, preemption, and graceful degradation across radio and edge resources. It should also determine when common information can be broadcast or cached, when individualized state requires protected capacity, and when a new session should be rejected before it becomes unsafe to depend on. PC5 and Uu coordination must respect their separate resource pools while deciding which object stays local, traverses the edge, or is duplicated. Evaluation should report deadline-qualified availability under a declared workload mix, complete-object goodput, resource cost, fairness across radio conditions, and the service degradation applied. Multiple independently scheduled UEs are essential for studying fleet contention; logical clients behind one UE cannot answer that question.

6.3 Jointly Adapt Information, Computation, and Communication

The open question is which representation and compute location can produce a useful result before its deadline when the original object cannot be delivered. A consumer may need a hazard flag, selected objects, uncertainty, a local occupancy region, an intended maneuver, or a trajectory instead of every sensor byte. The correct representation depends on the decision, deadline, path state, competing demand, and available vehicle or edge computation. Likewise, placing computation at an intersection helps only when the infrastructure has timely observations, suitable models, sufficient compute, and authority to produce the requested result.

Research should therefore co-design semantic reduction, region-of-interest selection, uncertainty-preserving compression, progressive transmission, early-exit inference, and computation placement. It should distinguish common information that can be published once from per-vehicle state that requires fresh correlation and authorization. For infrastructure-assisted intersections, this work must preserve the authority and safety constraints of the traffic-signal controller: an edge service may supply observations or advisories, but it must not bypass controller-authorized state or safety logic. The outcome should be evaluated with a task-specific decision metric: whether a correct action or safe deferral, with stated uncertainty and provenance, is available before its deadline. Evaluation should also report the associated privacy and energy cost and the time consumed by inference, serialization, security, and fusion. A smaller object is useful only if the receiver can tell what information was omitted and can still make the intended decision safely.

6.4 Build Service Continuity and Safe Degradation into Stateful Communication

The open question is how a multi-step service should preserve or safely terminate its state when a path or service component fails. Low RTT among completed packets does not preserve a multi-step service when a radio path, broker, receiver, edge process, backhaul, or remote assistant becomes unavailable. In addition, a late trajectory, duplicate command, or response from an expired session may be worse than an explicit rejection. Future CAVs will also cross roadside-unit, cell, edge, operator, and administrative boundaries. Thus, the service state and authority must either survive those transitions or fail early enough for local control to take over.

This problem requires session migration, replicated edge state, operations that remain safe if repeated, duplicate suppression, explicit terminal states, and authenticated authority transfer. Predicted interruption can trigger migration before the current path fails, caching, representation reduction, or a safe local mode, but the prediction must expose calibrated uncertainty. Multi-radio or multi-operator redundancy should be judged against correlated failures, duplicated load, and state inconsistency. Most importantly, communication support must integrate with the vehicle's minimal-risk behavior rather than replace it, consistent with automated-driving safety guidance [32]. Evaluation should measure detection time, deadline misses, uninterrupted-service fraction, recovery gap, state consistency, fallback completion, and safety outcome. The desired result is a service that either continues operating safely through a fault or enters a defined safe state. Its state, authority, deadline, and fallback must remain explicit through disruption.

6.5 Establish Benchmarks for Application-Ready CAV Communication

The open question is how to compare whether communication supports an application rather than only a radio or packet metric. The community needs benchmarks that retain both application meaning and experimental control. A byte sweep is reproducible but does not identify what the bytes must accomplish. Conversely, an end-to-end driving demonstration may hide which communication condition caused a failure. A useful portfolio should include compact periodic broadcast, many-to-many awareness, individualized request/reply, strict event-driven coordination, processed-perception objects, sustained uplink-heavy sessions, asynchronous transfer, and injected service interruption. Each workload should publish its complete object, direction, temporal pattern, delivery scope, deadline, freshness, required

availability, session behavior, and consequence of a missed result.

The benchmark portfolio should provide modular scenarios that separately and, where justified, jointly vary multiple independently scheduled UEs, radio condition, mobility and handover or roaming, interference, backhaul and edge distance, TDD context, and repeatable failure events. Each scenario should publish which factors are held fixed, the foreground and achieved background workload, and the causal scope of its evidence. Application and transport records are the minimum reproducibility artifacts. When a deployment can export network-side instrumentation, those records should be aligned with serving cell, Physical Resource Block use, Modulation and Coding Scheme, Block Error Rate, retransmission, bearer, and scheduler evidence. Otherwise, the report should state that radio-layer causal attribution is unavailable. Reports should include all-attempt deadline availability, completed-result latency, complete-object goodput, fragmentation overhead, resource cost, per-vehicle fairness, interruption gaps, and confidence bounds appropriate to the reliability claim. Open workload generators, object manifests, scenario definitions, and privacy-preserving traces would then let vehicle developers, agencies, equipment vendors, carriers, and researchers compare PC5, Uu, multi-connectivity, edge placement, schedulers, and future radio proposals by the application results they complete in time.

The central opportunity is to treat CAV communication as a portfolio of time-bounded services rather than a collection of packets. The International Mobile Telecommunications framework for 2030 and beyond supplies a broad future-network context [25], but a technology generation label does not establish application readiness. Progress instead requires an enforceable application/network contract, mixed-service resource management, joint adaptation of information, computation, and communication, explicit continuity and safe degradation, and reproducible application-level evidence.

7 Conclusion

Connected and automated vehicles depend on information and assistance beyond onboard sensing and computation. Yet the presence of V2X and 5G does not establish that a complete service will finish while it remains useful. To make this readiness question measurable, we design and implement IPI, a typed application protocol and reference interface that places selected J2735 message profiles and pre-encoded payloads beside a correlated CAV service-operation model. We then evaluate necessary communication conditions using separate J2735 functional checks, PC5 custom request/reply probes, and 5G IPI request/accepted-ACK probes on a real autonomous vehicle. These probes measure path response, not semantic execution of a remote CAV service.

Together, the measurements reveal complementary but incomplete operating envelopes. The measured direct V2X path completes compact local exchanges under favorable conditions, but its dependable envelope narrows as payloads grow or the field path weakens. Conversely, the measured 5G path carries larger objects, but tight deadlines and shared demand reduce what remains usable. Therefore, neither path, when used independently and as-is, turns connectivity into an application guarantee. Application-ready communication must ultimately be designed and judged by whether it completes the right service within its useful time and operating conditions.

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